VU MEDICAL ZONE

AMAAN KHAN

ANIMAL BEHAVIOR

ZOO 403 MIDTERM PPT

LECTURE 1 TO 40

Animal Behavior: Basics

Animal Behavior: Basics

Animal Behavior

"Sum of responses made by an animal to the external and internal stimuli"

Based on:

- Animal's physiological systems and processes

Animal Behavior: Basics

Ethology

"Scientific study of animal behavior in natural environment"

Animal Behavior: Basics

Examples of Behavior

Simple behaviors:

- Moving towards or away from light
- Blinking of eyes
- Reflex movements

Animal Behavior: Basics

Examples of Behavior

Complex behaviors

- Territory defence
- Courtship and mating
- Liking, disliking, fighting or caring
- Migration and navigation
- Building hives, nests, dens
- Producing voices, songs
- Communicating
- Learning, memorizing

Animal Behavior: Basics

Natural Selection in Evolution of Behavior

- Behaviors are essential for survival and reproduction
- Evolution of complex behavioral patterns is subject to natural selection

Animal Behavior: Basics

Natural Selection in Evolution of Behavior

Natural selection contributed to:

- Anatomical development of nervous system
- Enhance physiological features

Animal Behavior: Basics

Objectives of Ethology

Involves understanding of:

- Stimuli that elicit behavior
- Physiological mechanisms mediating response
- Animal's experiences that influence behavior
- Role of behavior in survival & reproduction
- Evolutionary history of behavior

END

Animal Behavior

Scientific Approaches to Study Animal Behavior

Scientific Approaches to Study Animal Behavior

Understanding Animal Behavior

Ethologists developed two complementary approaches:

- Neuroethological approach
- Ethological approach

Scientific Approaches to Study Animal Behavior

Neuroethological Approach

Involves:

- Bringing the animal into laboratory
- Observing its behavior in a simplified set of circumstances

Scientific Approaches to Study Animal Behavior

Simplified Set of Circumstances

In the laboratory:

- There are no predators
- Number and sex of conspecifics are controlled
- There are unusual lights, smells and sounds
- Animal is limited to a relatively small, confined space

Scientific Approaches to Study Animal Behavior

Neuroethological Study of Behavior

This approach focuses on:

- Study of neuronal responses/signals generated during the manifestation of an animal's behavior

Scientific Approaches to Study Animal Behavior

Neuroethological Study of Behavior

- Nervous system of animal is exposed surgically
- Recording from neurons as behavior is going on

Scientific Approaches to Study Animal Behavior

Neuroethological Approach—Significance

- Studies under restricted conditions are useful for obtaining neuronal information about behavior
- Answering physiological queries about behavior

Scientific Approaches to Study Animal Behavior

Neuroethological Approach—Limitations

- Such experiments cannot equate an animal's dealing in their everyday lives in natural environment

Scientific Approaches to Study Animal Behavior

Ethological Approach

Involves:

- Going into the field
- Observe animal's behavior in its natural environment

Scientific Approaches to Study Animal Behavior

Limitations of Ethological Approach

- For a physiologist, natural conditions render experimental constraints
- It is impossible to record activity from animal's nervous system that is involved in producing behavior

END

Animal Behavior

Proximate Causes of Behavior

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Proximate Causes

- The word 'proximal' comes from the Latin word 'proximus', which means 'close' or 'near'.
- Therefore, proximal influences to our behaviors are influences from our immediate situation.

- Proximate factors may be defined as:

  “Those **external stimuli which initiate or maintain biological processes under most favourable ecological conditions**.”

- Proximate causation explains biological function in terms of immediate physiological or environmental factors.
- Example:
  - ✓ Eating to satisfy hunger is a physiological response
  - ✓ A female animal chooses to mate with a particular male during a mate choice trial. A possible proximate explanation states that one male produced a more intense signal, leading to elevated hormone levels in the female producing copulatory behaviour.

Difference between proximate and ultimate causes

1. Genetic–developmental mechanisms	1. Historical pathways leading to a current behavioral trait
Effects of heredity on behavior	Events occurring over evolution
Development of sensory–motor systems via gene–environment interactions	from the origin of the trait to the present
2. Sensory–motor mechanisms	2. Selective processes shaping the history of a behavioral trait
Nervous systems for the detection of environmental stimuli	Past and current usefulness of the behavior in promoting lifetime reproductive success
Hormone systems for adjusting responsiveness to environmental stimuli	
Skeletal–muscular systems for carrying out responses	

Sources: Holekamp and Sherman [532], Sherman [1046], and Tinbergen [1139]

Animal Behavior

Ultimate Causes of Behavior

Ultimate Causes

We ask questions like:

- Why animal behaves as it does?
- What is the evolutionary cause of behavior?

These questions explains the ultimate cause of behavior.

The ultimate level of causation refers to:

- the behavior's evolutionary significance
- how the behavior enhances reproductive fitness

- Ultimate causation explains traits in terms of evolutionary forces acting on them.

Example:

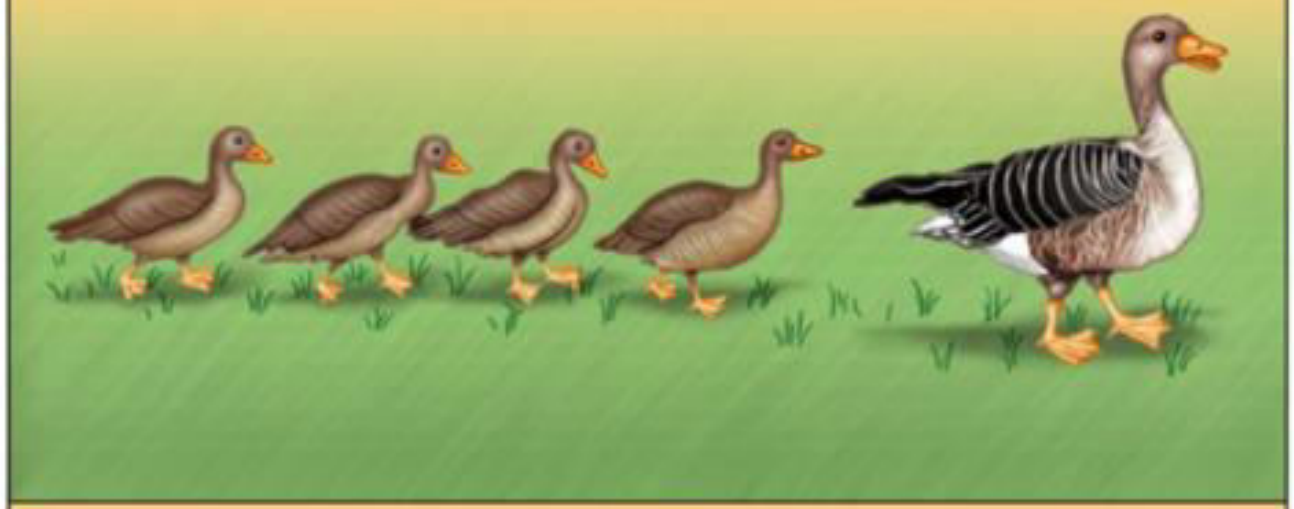
Female animals often display preferences among male display traits, such as song. An ultimate explanation based on sexual selection states that females who display preferences have more vigorous or more attractive male offspring.

Difference between proximate and ultimate causes

1. Genetic–developmental mechanisms	1. Historical pathways leading to a current behavioral trait
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BEHAVIOR: Young geese follow and imprint on their mother.



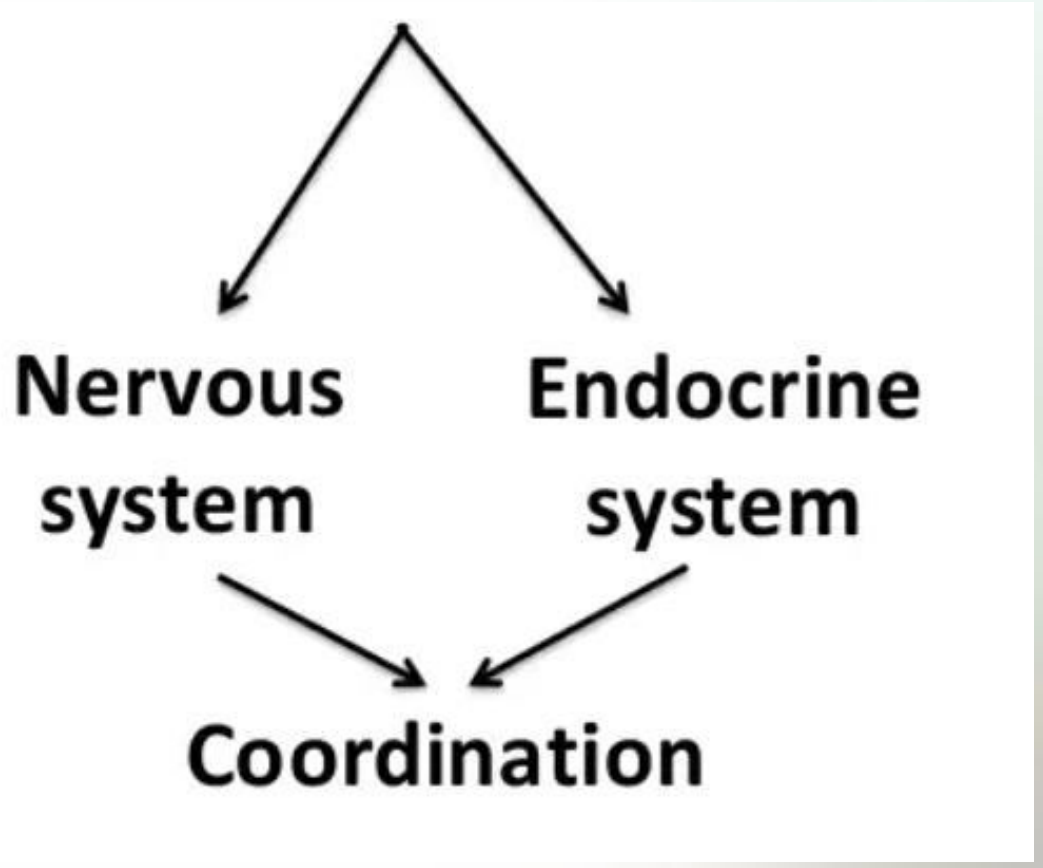
PROXIMATE CAUSE: During an early, critical developmental stage, the young geese observe their mother moving away from them and calling.

ULTIMATE CAUSE: On average, geese that follow and imprint on their mother receive more care and learn necessary skills, and thus have a greater chance of surviving than those that do not follow their mother.

Animal Behavior

Nervous System and Behaviour 1

Behaviour is sum of two systems



Behaviour is sum of two systems

Biologically, behavior is a response to a stimulus shown by the collective mechanism of Nervous and chemical coordinations according to a given situation.

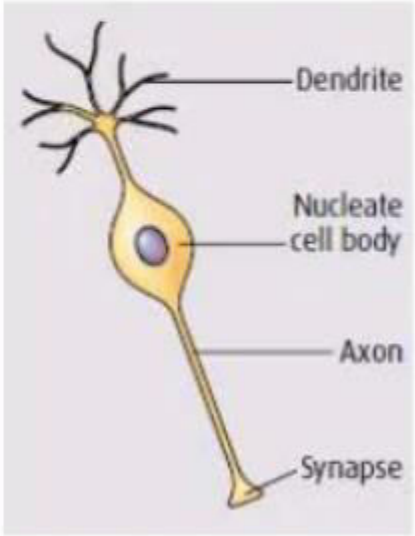
Nervous system and Behaviour

Components of Nervous system

- Main building blocks and units of nervous system is

"Neuron"

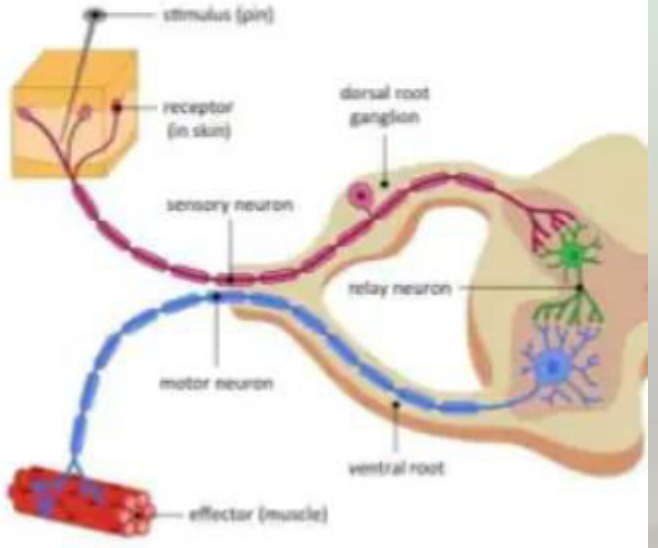
- Have three components
  1. Cell body
  2. Axon (Move the nerve impulse away from cell body)
  3. Dendron (Move the nerve impulse towards the cell body)



Nervous system and Behaviour

Classes of neurons

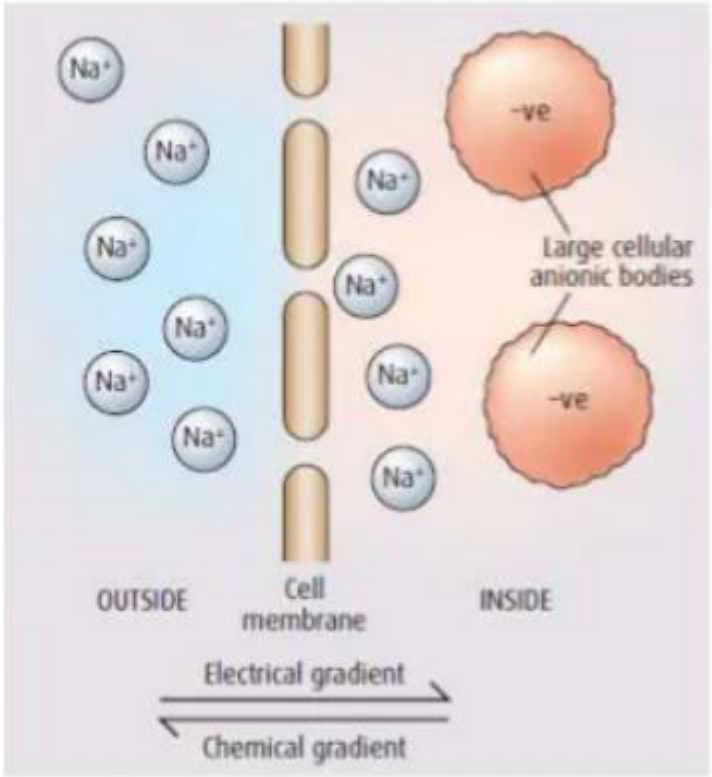
- Three main classes of neurons
  - Sensory neurons
  - Interneurons (relay neuron)
  - Motor neurons
  - "messages through nerve impulse pass in the form of **"Nerve Impulse"**
  - Nerve impulse is an electrochemical change which runs along the length of neuron.



Nervous system and Behaviour

Nerve Impulse

- The information handled by neurons takes the form of electrical signals.
- These are propagated as small changes in voltage between the inside and outside of the cell.
- To facilitate this the cell membranes of neurons are permeable to ion flow i.e. Na^+ & K^+.

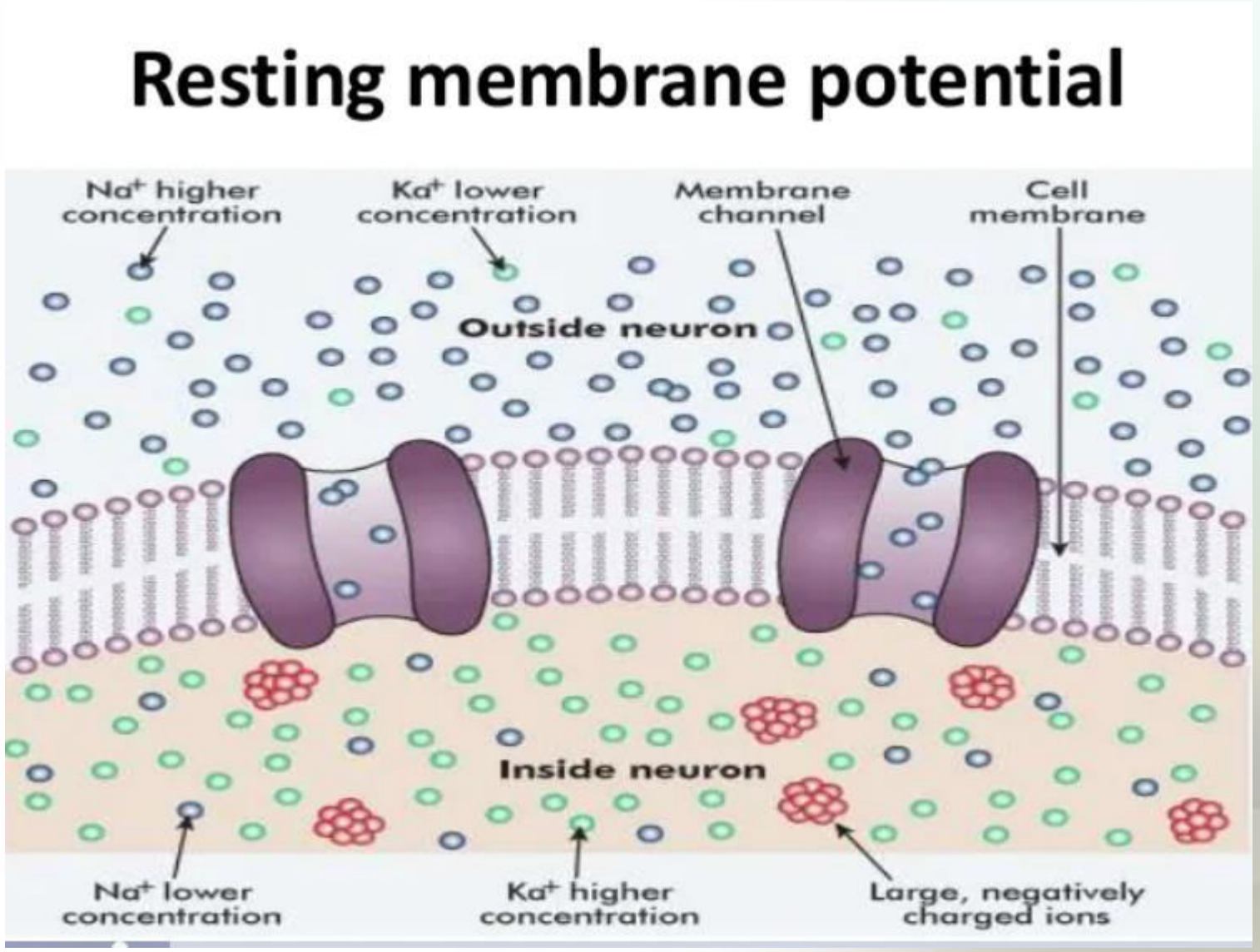


Nervous system and Behaviour

Membrane Potential: Stored electrical charge of membrane to conduct a nerve impulse.

- Resting membrane potential
  - Two different (Na^+ & K^+) gradients of ion movement across this membrane exist.
  - Ions move down the concentration gradient across the cell membrane, and result in similar gradient one both sides.
  - But membrane is more negatively charged inside due to presence of negatively charged macromolecules i.e. proteins.
  - Na^+, K^+, Ca^{2+} and Cl^- ions have a balanced diffusion gradient across cell membrane known as **equilibrium membrane potential** or **resting membrane potential.**
  - At this stage, membrane is more negatively charged inside than outside i.e. -70mV.

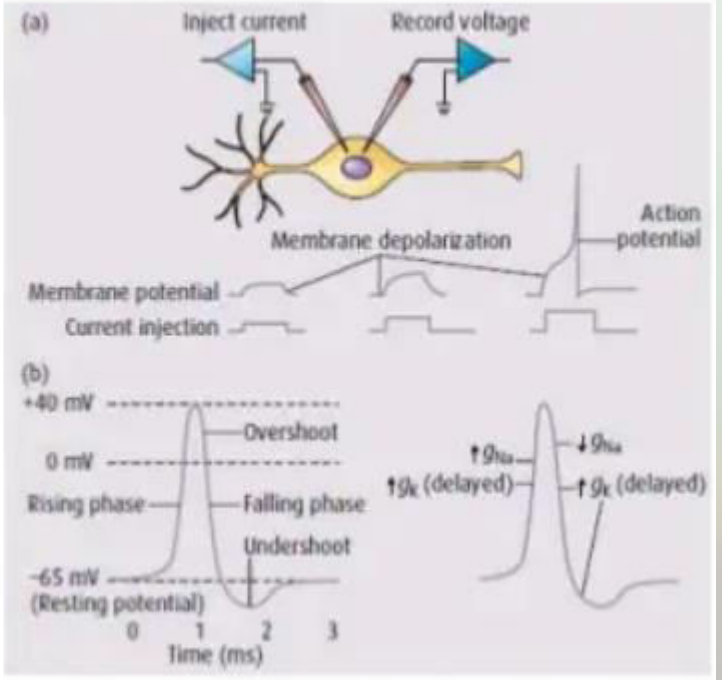
Nervous system and Behaviour



Nervous system and Behaviour

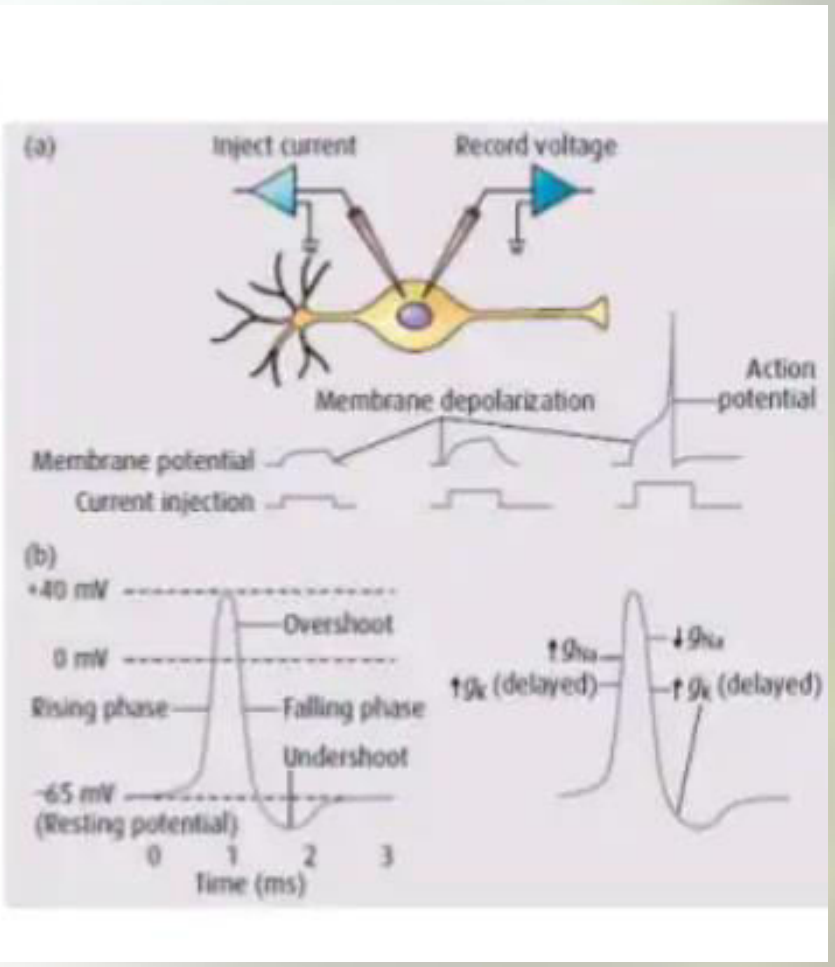
Action Potential

- If the interior of a neuron is made to have a greater positive charge than the resting potential the membrane will depolarize.
- Depolarizations tend to be excitatory in nature, stimulating activity in the neuron, whereby known as **Action Potential**.
- If the charge change related to a depolarization is sufficiently large, a very particular electrical phenomenon called a **spike** will be triggered.
- Action potential is "all or none response".



Hyperpolarization

- The application of an increased negative charge within the cell will result in membrane hyperpolarization.
- Hyperpolarizations tend to be inhibitory.

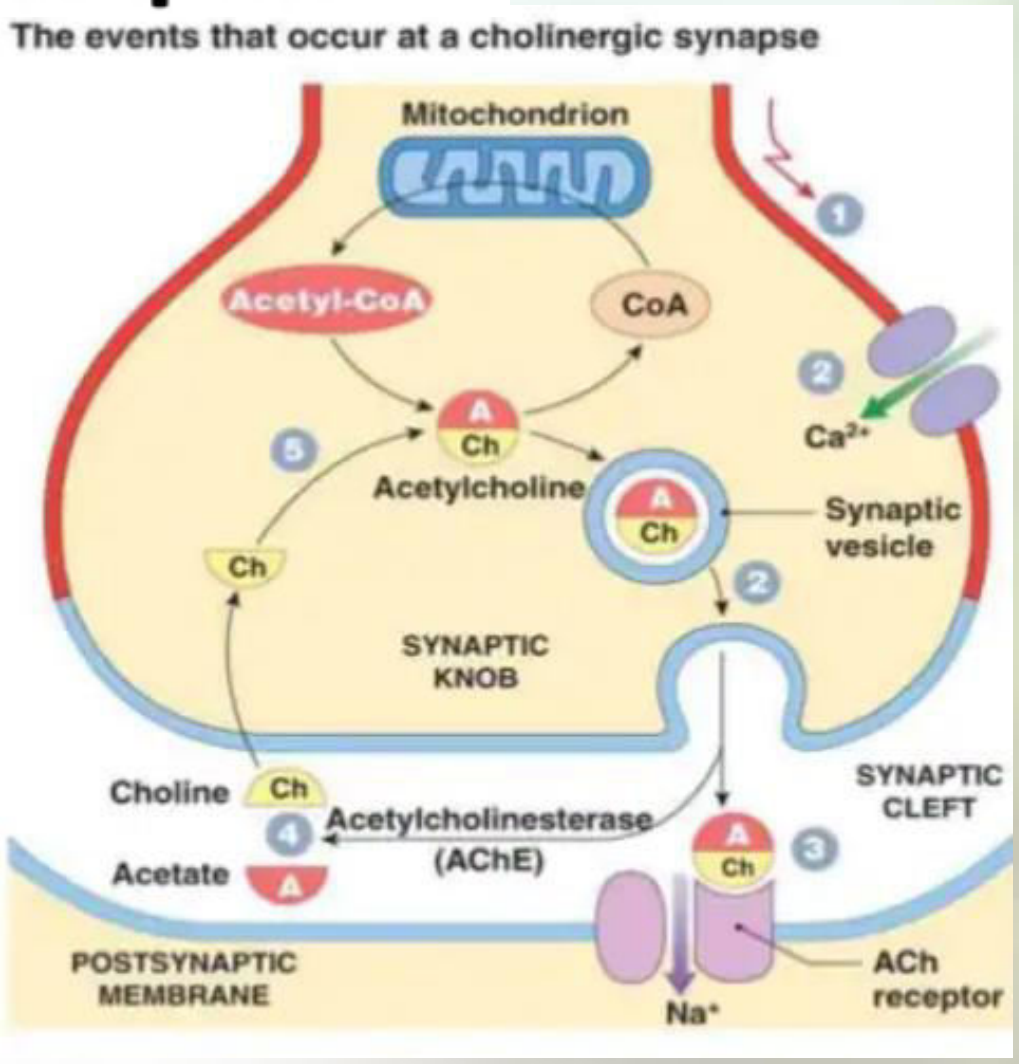


Nervous system and Behaviour

Synapse

Events Occurring at Synapse

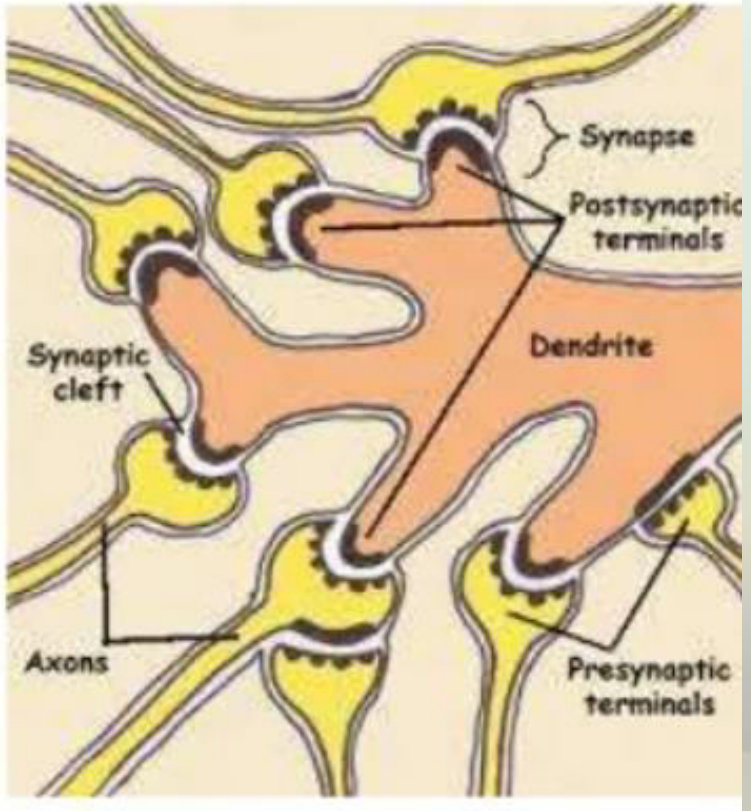
1. An arriving action potential depolarizes the synaptic knob.
2. Calcium ions enter the cytoplasm, and after a brief delay, ACh is released through the exocytosis of synaptic vesicles.
3. ACh binds to sodium channel receptors on the postsynaptic membrane, producing a graded depolarization.
4. Depolarization ends as ACh is broken down into acetate and choline by AChE.
5. The synaptic knob reabsorbs choline from the synaptic cleft and uses it to synthesize new molecules of ACh.



Nervous system and Behaviour

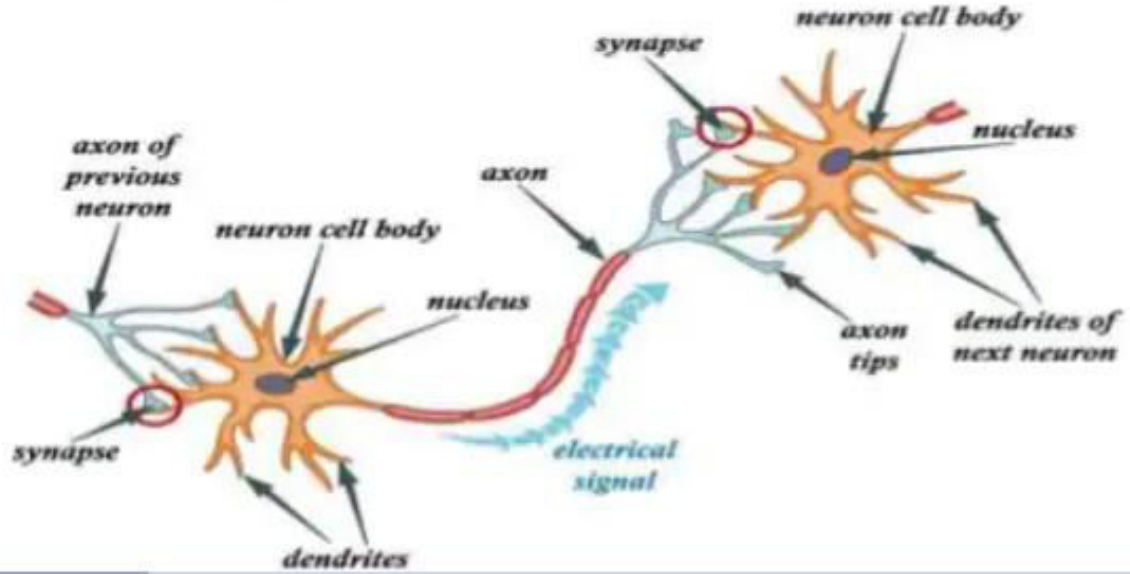
Potential required to initiate the Impulse

- Sometimes a single postsynaptic potential (PSP) is enough to induce a spike.
- Often the combined effect of a number of them is required.
- Some nerve cells exhibit a phenomenon termed **summation** whereby a series of small PSPs coming from one or a number of sources have an additive effect.



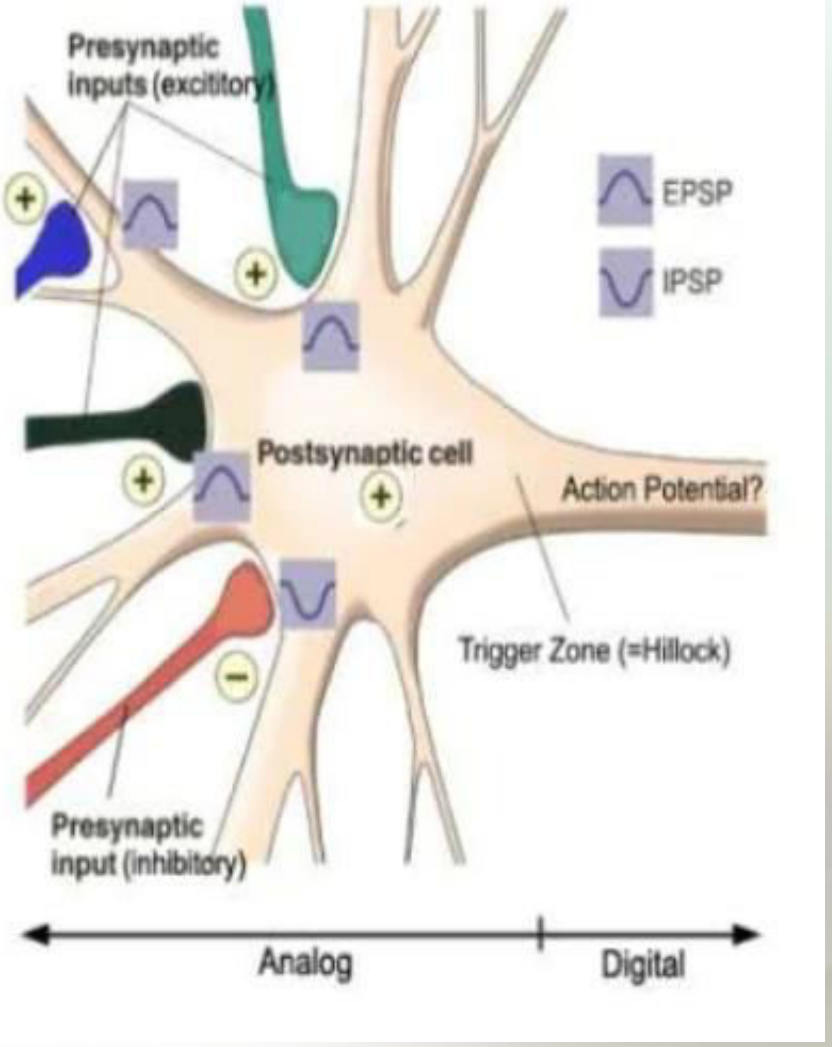
Nervous system and Behaviour

- Other cells require a series of PSPs that rather than being added together each elicit a slightly larger PSP than themselves until the action potential threshold is reached. This phenomenon is termed **synaptic facilitation.**
- Once a nerve impulse has started, it does not stop until the synapse of the next axon reaches.



Nervous system and Behaviour

- The pre-synaptic potential may be Excitatory pre-synaptic potential (**EPSP**) which will depolarize the post synaptic dendron.
- Or it can be Inhibitory pre-synaptic potential (**IPSP**) which will hyperpolarize the post synaptic dendron.



Integration

- How likely it is that a spike in a presynaptic cell will stimulate a spike in the postsynaptic cell with which it communicates will depend upon the precise combination of EPSPs and IPSPs involved – a process termed **integration**.

Animal Behavior

Nervous System and Behaviour 2 (Examples)

Behaviour is sum of two systems

Examples of Animal Behavior and involvement of Nervous system

Behaviour is sum of two systems

- To understand the relation of nervous system with animal behavior, we will consider two examples:
  - **Controlling the prey capture (Toad)**
  - **Controlling escape behavior (Cray fish)**

Nervous system and Behaviour

Controlling the prey capture in toads

Nervous system and Behaviour

- It was done by Jörg-Peter Ewart and his coworkers.
- They said that **the particular area of nervous system that is responsible for prey capture behavior in toads is called as FEATURE DETECTOR.**
- According to them, during capturing prey, following steps are followed:
  - The toad moves so that it faces the prey animal.
  - The toad approaches the prey animal to within striking distance.
  - The toad uses its tongue to strike at the prey animal and captures it.
  - The toad swallows the animal.
  - The toad wipes its mouth.





Nervous system and Behaviour

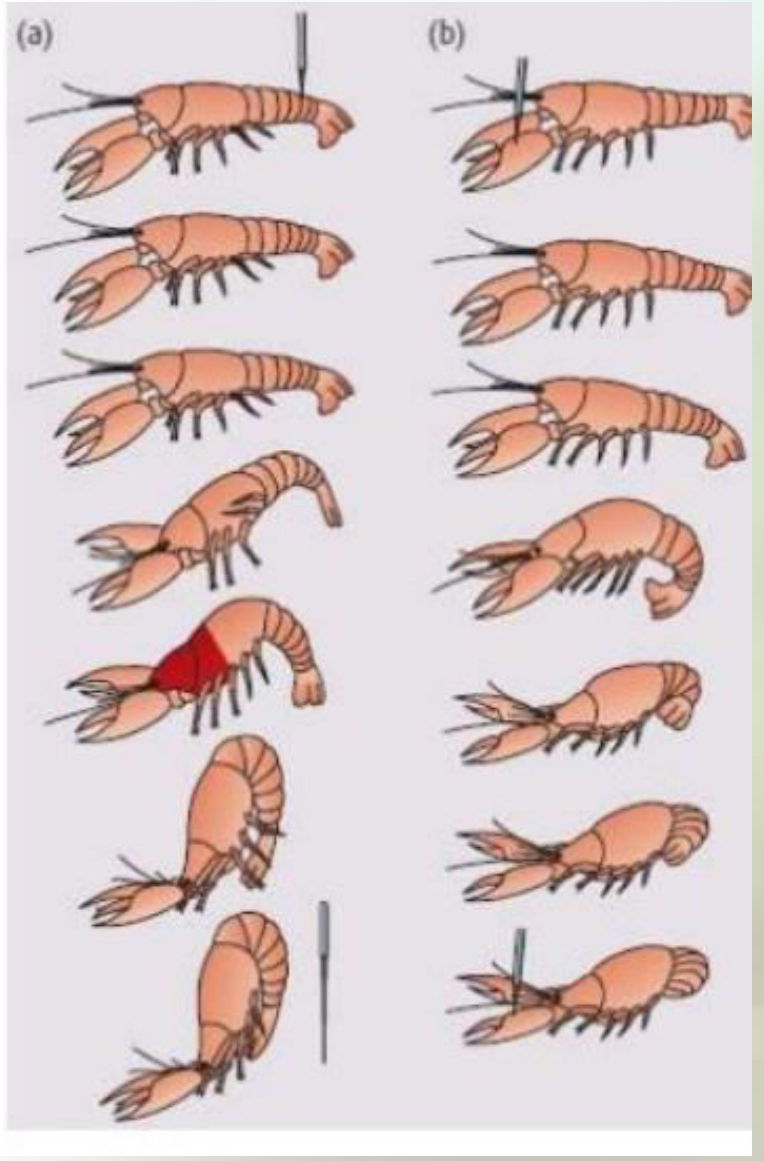
- According to them, this behavior of frog is instinctive reflex.
- **Instinct** is any behavior that is innate by its nature i.e. present in an organism by birth.
- They said it is instinctive because according to them"**a newly metamorphosed toad that is completely prey-naïve will perform the whole sequence**"
- Step 1 & 2 can be skipped if the prey is in the very near vicinity.
- So if a toad is treated with a "**model worm**" the nervous coordination of this whole phenomenon can be studied.

Nervous system and Behaviour

Controlling the escape behavior

Nervous system and Behaviour

- The figure illustrates the rapid escape behavior, the tail-flip, of the crayfish.
- Flipping the highly muscular tail propels the animal away from the source of danger, and as one would perhaps expect the onset of the behavior following stimulation is extremely rapid.
- It has a latency (a time period of showing a response) of as little as 10 milliseconds.



Nervous system and Behaviour

The Nervous Mechanism of tail flip

- Tail flip is caused by rapid contraction and extension of the massive abdominal muscles.
- This activity is controlled by
  - **Large motor neurons** on both sides of the body, coming from Brain.
  - **Motor giants** which sends nerve impulse to every involved muscle fiber.
  - All neurons are interconnected by a number of **Giant Interneurons (GIs)**. GIs are of two types:
    - Medial Giant Interneurons (MGIs): Starting from the brain and all the way into the tail.
    - Lateral giant interneurons (LGIs): Have a very different structure. These interneurons have a chain of cell bodies in every segment.

Animal Behavior

Neurosensory Basis of Behavior

Neurosensory Basis of Behavior

Role of Nervous System

- NS plays essential role in animal behavior
- Animals manifest behaviors in response to environmental stimuli
- Stimuli act as sensory input to NS
- NS produces motor output
- Motor output controls contraction of muscles which generate all behavioral acts

Neurosensory Basis of Behavior

Neurosensory Basis of Behavior

- Virtually all animal behaviors have neurosensory basis
- Rooted in vast array of neural circuitry (interconnecting circuits of neurons)
- Laying b/w afferent sensory pathways and efferent motor pathways

Neurosensory Basis of Behavior

Neural Circuitry

- Lower animals have simple neural network
- They show simple behavioral patterns
- Neural circuitry is more complex in animals with a CNS

Neurosensory Basis of Behavior

Neural Circuitry

- Increasing number of interneurons in CNS, increases behavioral potential of an animal

Neurosensory Basis of Behavior

Complex Behaviors

- Higher animals with complex CNS show complex behaviors:
  - Learning from experience
  - Memory
  - Associating combinations of stimuli

Neurosensory Basis of Behavior

Complex Behaviors

- Are less predictable—than simple predictable reflex behaviors
- Depend on information stored from past experiences

END

Animal Behavior

Neuronal Plasticity

Neuronal Plasticity

Neuronal Plasticity

- “Capacity of nervous system to change with experience”
- Basic architecture of nervous system establishes during embryonic development
- Behavioral patterns modify with time, age and experience
- Premier importance for fitness and survival

Neuronal Plasticity

is the basis of

Examples of Neural Plasticity

- Learning, memory
- Development of motor skills and habits
- Ability to respond to stimuli in diverse ways (different from fixed, programmed reflexes)
- Lies behind human intelligence

Neuronal Plasticity

Mechanism of Neuronal Plasticity

- The reshaping (plasticity) of nervous system occurs at synapses
- Often referred as synaptic plasticity
- Mechanisms underlying synaptic plasticity are currently under intense research

Neuronal Plasticity

involves

Synaptic Plasticity

- Modifying synapses over the course of lifetime due to developmental events
- Synaptic connections established in embryos are refined into adult patterns (mature synapses)
- Synapse maturing results in synaptic strengths enabling learning and memory

Neuronal Plasticity

Synaptic Plasticity

- Changes for learning and memory depend on a retrograde (reverse) signal
- Sent from postsynaptic neuron to presynaptic neuron

Neuronal Plasticity

Mechanisms of Synaptic Plasticity

Occurs by two types of mechanisms:

- Presynaptic mechanisms
- Postsynaptic mechanisms

Neuronal Plasticity

Presynaptic Mechanisms

- Involve changes in the functioning of presynaptic terminals

Example

- A change in the amount of transmitter released from presynaptic terminals

Neuronal Plasticity

Postsynaptic Mechanisms

- Involve changes in postsynaptic neurons

Example

- A change in postsynaptic apparatus resulting in altered frequency of APs

END

Physiology of Behaviour in changed Environment 1

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Physiology of Behaviour in changed Environment

- In physiology we study the functions and mechanisms in a living system. It focuses on how organisms, organ systems, individual organs, cells, and biomolecules carry out the chemical and physical functions in a living system.

Physiology of Behaviour in changed Environment

- Animals depend on their physical features to help them obtain food, keep safe, build homes, withstand weather, and attract mates.
- These physical features makes it possible for the animal to live in a particular place and in a particular way.

Physiology of Behaviour in changed Environment

Due to the stimulus present animal can show his response in form of:

1. Morphological Responses
2. Behavioral Responses
3. Physiological Responses

Physiology of Behaviour in changed Environment

1. Morphological Responses

- changing in the of the body appearance

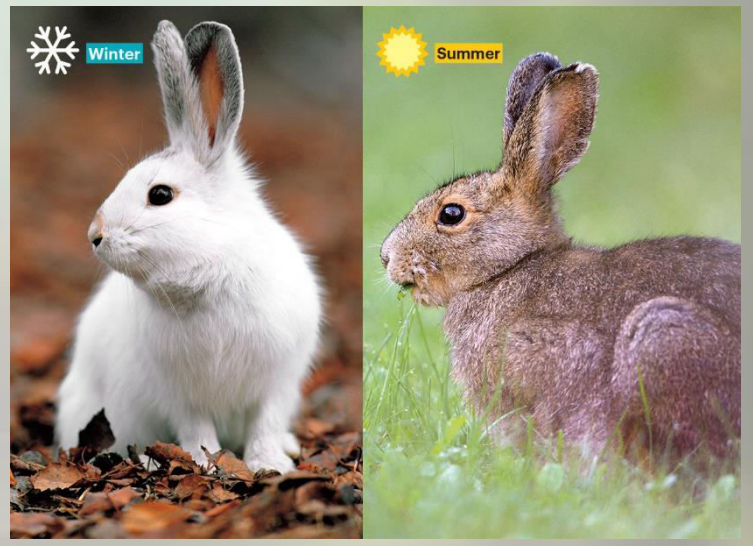
- growing thicker fur or change in fur color in winter

Examples:

- chameleon change colour when it feel threatened.
- Arctic hare change its coat colour from blue-grey to white for better camouflage during winters and turn again to blue-grey in spring.



Chameleon Changing Colour



Arctic Hare

Physiology of Behaviour in changed Environment

2. Behavioral Responses

- changing behavior to adapt to the change
- moving to a more favorable location
- cooperative behavior
- agonistic behavior when threatened

Examples:

- During the winter season, Houbara bustard finds refuge in plain lands of Sindh, Southern Punjab and Balochistan.



Houbara bustard

Examples:

- “Cleaner” fish swim into the mouths of the bigger fish—called “clients”—in order to eat parasites and harmful bacteria. The cleaners get a meal, and the clients get a healthier mouth.





“Cleaner” Fish

Examples:

- In lions, males cooperate to defend the pride, and females hunt and share the kill with other pride members.



Lion’s Pride Hunting

Physiology of Behaviour in changed Environment 2

Physiology of Behaviour in changed Environment

3. Physiological Responses

- Changing the functioning of the body
- Acclimation (dilating capillaries to release heat)
- Acclimatization

Physiology of Behaviour in changed Environment



Colour changing in Animals

- Rapid colour change may occur due to various “triggers” including temperature or light (a reflexive response via light-sensitive receptors in skin)
- Most importantly, animals change colour in response to their surroundings including variations in background colour, presence of predators, mates or rivals.

Physiology of Behaviour in changed Environment



Sweating

When the body temperature rises, the sympathetic nervous system stimulates the sweat glands to secrete water to the skin surface, where it cools the body by evaporation. Animals such as horses, monkeys, apes, and hippos sweat etc. use sweating to maintain body temperature.

Physiology of Behaviour in changed Environment



Panting

When an animal pants, it evaporates water from internal body structures. Panting expels hot air and brings in cool air. This physiological response is used by many mammals, most birds, and some reptiles to cope up with the hot environment.

Physiology of Behaviour in changed Environment



Shivering

It is also a physiological response against cold environmental conditions. When an animal shiver its body muscles tighten and relax in rapid succession. This involuntary muscle movement is body's natural response against cold and trying to warm up.

Physiology of Behaviour in changed Environment

Acclimatization

An adaptive change within an individual animal.

Happens due to chronic exposure to new, naturally occurring environmental conditions.

Example

An animal migrates to a high altitude area faces low oxygen partial pressure

Effects

- Lung ventilation rate increases initially
- Ventilation drops to normal in few days
- Animal has acclimatized to the high-altitude

Physiology of Behaviour in changed Environment

Acclimation

Changes are induced experimentally in the laboratory

Example

Simulating high-altitude conditions in lab.

An animal physiologist places the animal in a hypobaric chamber.

Effects

Breathing rate reacts in the same way.

Animal becomes acclimated to the experimental conditions.

Physiology of Behaviour in changed Environment

Both of these are:

- Acquired characters
- Restricted to one or few members of species
- Not inheritable
- Have no evolutionary significance
- Reversible

Hormones and Animal Behaviour 1

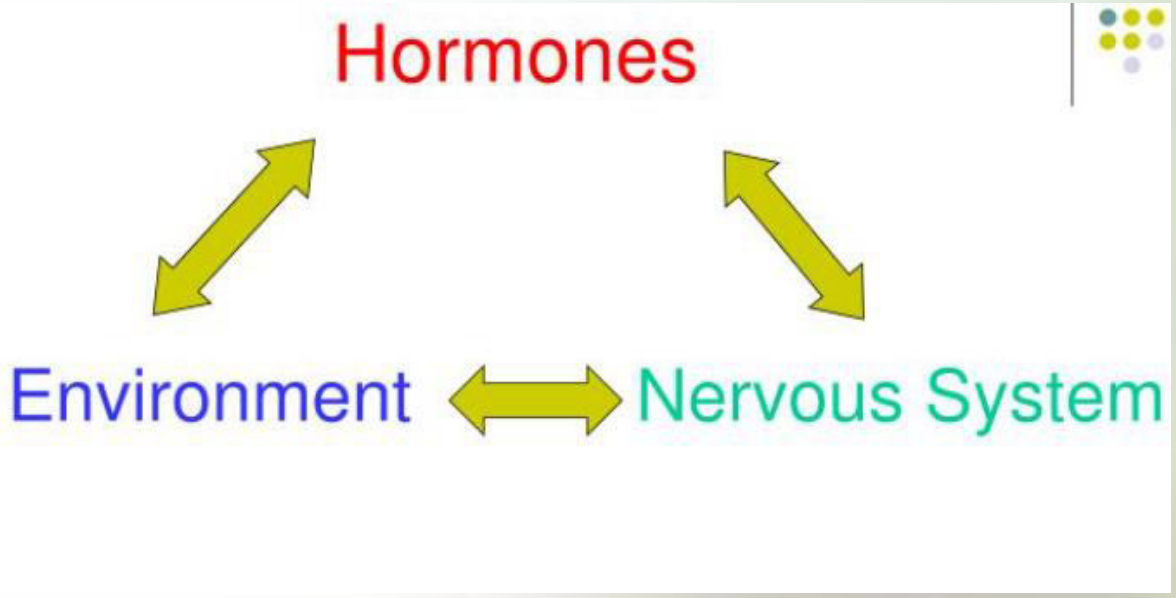
Hormones and Animal Behaviour

The scientific study of the interaction between hormones and behavior is called **behavioral endocrinology.**

Hormones and Animal Behaviour

This interaction is bidirectional: hormones can influence behavior, and behavior can sometimes influence hormone concentrations.

Hormones and Animal Behaviour



Hormones and Animal Behaviour 2

Hormones and Animal Behaviour

Hormones are chemical messengers

Released from endocrine glands

Travel through the blood system to influence the nervous system to regulate behaviors such as:

- Aggression
- Mating
- parenting of individuals

Hormones and Animal Behaviour

Examples of hormones that influence behavior:

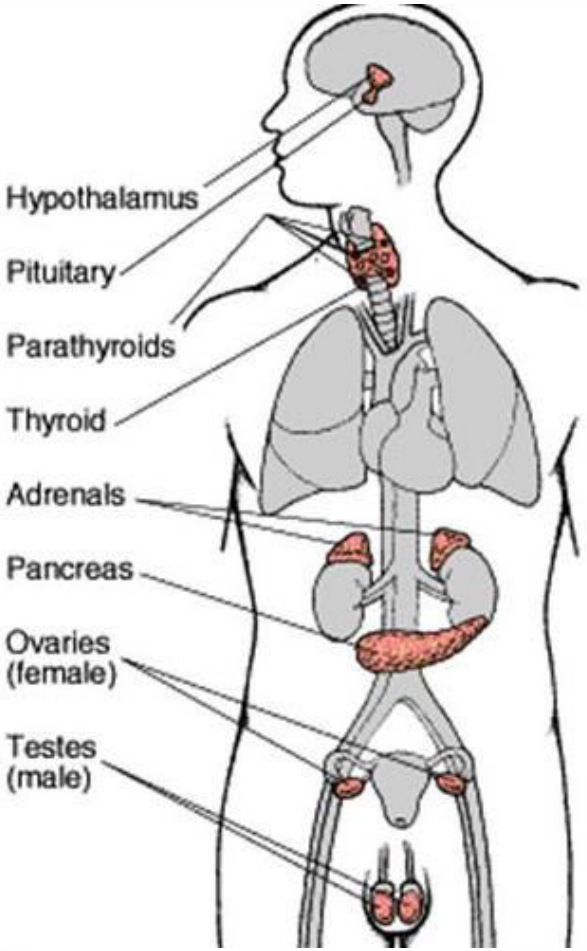
- Steroid hormones such as **testosterone**
- Estradiol (a common type of estrogen)
- Progesterone (a common type of **progestin**)
- Cortisol (a common type of glucocorticoid)

Hormones and Animal Behaviour

Several types of protein or peptide (small protein) hormones also influence behavior, including

- Oxytocin
- Vasopressin
- Prolactin
- Leptin

Hormones and Animal Behaviour



The Endocrine Glands

These are the glands that release the particular hormones.

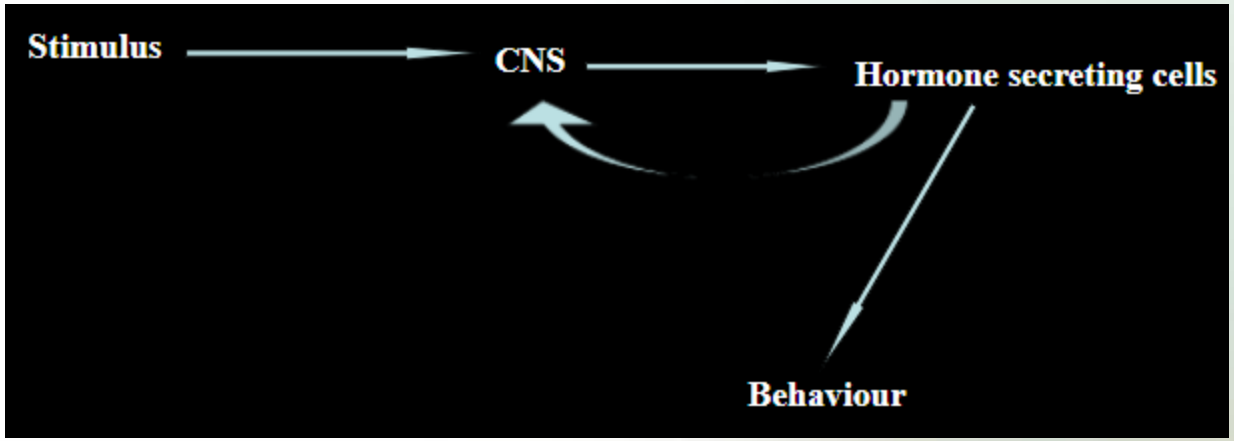
Hormones and Animal Behaviour

Endocrine Glands - What do they do

Gland	Hormone	Effect
Pineal	Melatonin	Annual reproductive cycle
Posterior Pituitary	Oxytocin, vasopressin	Milk ejection, birth, water balance
Anterior pituitary	Luteinizing hormone (LH)	Corpora lutea formation Progesterone secretion Androgen secretion
	Follicle stumulating hormone	Follice development Ovulation (with LH and estrogen.) Spermatogenesis
	Prolactin	Milk secretion Parental behaviour (birds)
	Adrenocorticotropic hormone(ACTH)	Regulates adrenal glands
Intermediate pituitary	Melanophore stimulating hormone	Colour change
Adrenals(cortex)	Steroids	Water balance, metabolic rate, electrolyte balance, blood sugar, stress reaction
Adrenals (medulla)	Adrenaline, noradrenaline	Blood sugar, stress reaction
Pancreas	Insulin	Blood sugar
Testes	Androgens	Testis development, sexual activity, spematogenesis
Ovaries	Estrogens	Uterine growth, mammary gland development, sexual activity
	Progesterones	gestation

Hormones and Animal Behaviour

How effect is brought about



Hormones and Animal Behaviour 3

Hormones and Animal Behaviour

Examples of Hormones Control in Animal Behaviour

Anolis Lizards



Testosterone is critical for stimulating male sexual behaviors. Castration reduces the display of male reproductive behaviors, and testosterone can prevent or reverse that effect.

Hormones and Animal Behaviour

- Green Anoles use their pink dewlaps for two purposes.
  - ✓ For breeding, displaying their intent to females.
  - ✓ For establishing territory among adult males. By displaying his dewlap, the lizard can tell other males that they have entered his home range.

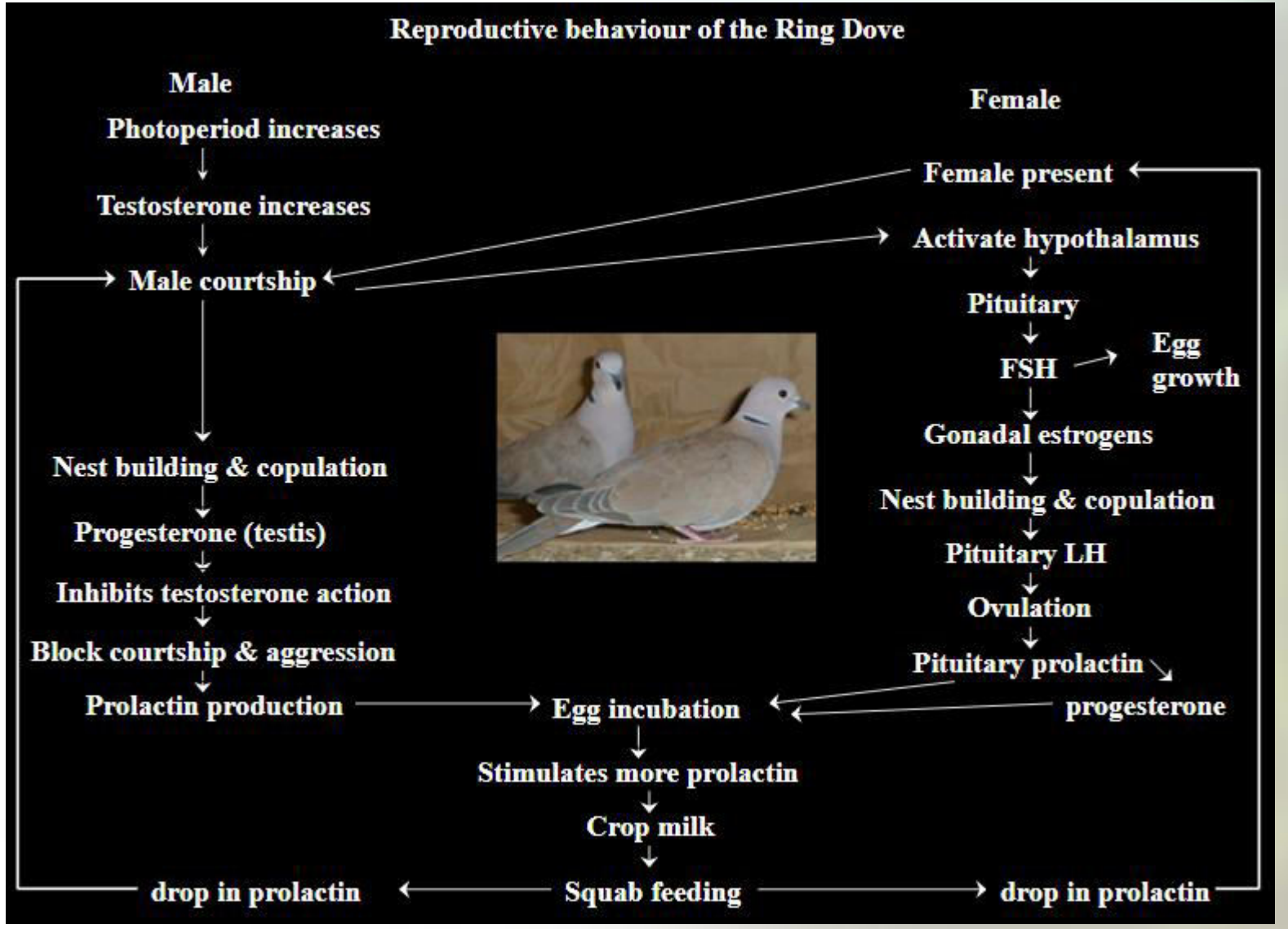
Hormones and Animal Behaviour

Reproductive Behaviour in Ring Doves

Male and female ring doves coordinate their participation in:

- ✓courtship
- ✓nest building
- ✓Incubation
- ✓care of young

Hormones and Animal Behaviour



Animal behaviour

Types of Animal behaviour

Types of Animal behaviour

There are 2 basic types of behaviour

1. Innate behaviors or Inherent behaviour
2. Learned behaviors or acquired behaviour

1. Nature/innate

Instinct or genes determine behaviour

2. Nurture/learned

Experience and learning influence behaviour

Innate behaviors or Inherent behaviour

1. **Innate behaviors**
   - automatic, fixed, “built-in”, no “learning curve”
   - despite different environments, all individuals exhibit the behavior
   - Examples: Early survival, reproduction, kinesis, taxis

Innate behaviors or Inherent behaviour

Examples:

1. Web making in spiders



2. Nest building in birds



3. Fighting among male stickleback fish



4. Cocoon spinning in insects



5. Swimming in dolphins



Types of Animal behaviour

2. **Learned behaviors**

- modified by experience
- variable, changeable
- flexible with a complex & changing environment

Types of Animal behaviour

1. By watching their mother, baby ducks learn how to avoid danger and to know what is good to eat. This is an example of learned behavior.



2. Prairie dogs learns by experience to only make alarm call s when they hear predators.



Animal Behavior

Inborn, Innate or Instinctive Behaviors

Inborn, Innate or Instinctive Behaviors

- **Innate behaviour** is inborn or innate sequence of activities which are predictable, species specific, genetically controlled and independent of past experience.
- Since this type of behaviour follows a fixed pattern (FAP) which is predictable and found in all members of a species it is also called stereotyped behaviour.

Inborn, Innate or Instinctive Behaviors

Innate or Inborn Behaviors

- Developmentally fixed
- Complex behaviors
- Exhibited by most members of a species
- Are built-in part of nervous system at the time of birth
- Some develop at an appropriate point in maturation

Inborn, Innate or Instinctive Behaviors

Innate or Inborn Behaviors

- Also known as innate, inherent or instinctive behaviors,
- Complex and different from reflexes (simple responses of an organism to a specific stimulus)

Inborn, Innate or Instinctive Behaviors

Characteristics of innate behaviour

- Pattern of behaviour is inherited. It is passed on from parents to offspring.
- It is unlearned behaviour.
- It occurs in all the members of a species hence it is species specific and predictable.
- It is not dependent on past experience as it is an inborn response to a stimulus
- It takes place in individuals even when kept in isolation away from their fellow members
- Innate behaviour has high adaptability and survival value.

Inborn, Innate or Instinctive Behaviors

Genetic Basis of Instinctive Behaviors

- Closely controlled by genes
- Not (or little) influenced by environment

Inborn, Innate or Instinctive Behaviors

Genetic Basis of Instinctive Behaviors

- Instinctive responses are determined by neural programs
- Neural programs are encoded by hereditary material
- Contain information for anatomical and physiological organization of NS

Inborn, Innate or Instinctive Behaviors

Exhibition of Instinctive Behaviors

- Don't have to be learned or practiced
- Are performed in a reasonably complete form even when exhibited first time (without prior exposure or experience)
- Can be modified through experience only to a limited extent

Inborn, Innate or Instinctive Behaviors

Types of innate behaviour

There are different types of innate or stereotyped behaviour

i. Taxis
ii. Irritability
iii. Kinesis
iv. Instincts
v. Reflexes
vi. Motivation

Inborn, Innate or Instinctive Behaviors

Examples of Instinctive Behaviors

- Human newborn’s suckling behavior
- Newly hatched sea turtles, automatically move from beach to the ocean
- A marsupial climbs into its mother's pouch just after birth

Inborn, Innate or Instinctive Behaviors

Examples of Instinctive Behaviors

- Fighting behaviors
- Courtship behaviors
- Building of nests
- Internal escape functions

Inborn, Innate or Instinctive Behaviors

Importance of Innate Behaviors

- Usually involve basic life functions important in survival
- Role varies from species to species
- Especially important for animals with short life spans and poorly developed CNS

Inborn, Innate or Instinctive Behaviors

Role of Instincts in Higher Animals

- Animals with more complex nervous systems rely more on social learning (due to developed cerebral cortex)
- Instincts play lesser role in them

END

Animal Behavior

Cocoon spinning in insects such as moths

Examples of Instinctive Behaviors

Cocoon spinning _ an examples of instinctive or innate behaviour

- A cocoon is a protective casing, usually made of silk.
- This casing protects the larval, or immature stage, of an insect from the elements, such as extreme temperatures.
- Inside the cocoon, the larva, which is often wormlike, will undergo metamorphosis and emerge as an adult

Cocoon spinning in insects such as moths

Cocoon-Building Insects

- Butterflies and moths are the most common cocoon building insects.
- Their larvae, which are **caterpillars**, are voracious eaters.
- Caterpillars spin silk, and this silk is used to form the cocoon for the pupal stage of development – the final stage before adulthood.

Cocoon spinning in insects such as moths

Cocoon spinning behaviour

a. Presence of an internal factor pressing for the initiation of spinning.

b. During spinning, turning the entire body around. The spinning behavior includes two distinct and mutually exclusive movement patterns (FAPs)

   i. the stretch-bend

   ii. the swing-swing

Cocoon spinning in insects such as moths

c. Periodically, the silkworm halts one movement pattern, turns Its entire body around, and begins the second series of movements.

d. During period spinning, the silkworm suddenly halts construction of the outer envelope of the cocoon. It then spins a loose, intermediate layer and, finally, a compact inner envelope.

e. If these actions are not elicited by the external environment, they must necessarily be the outcome of events originating within the insect itself.

End

Animal Behavior

Web making in spiders

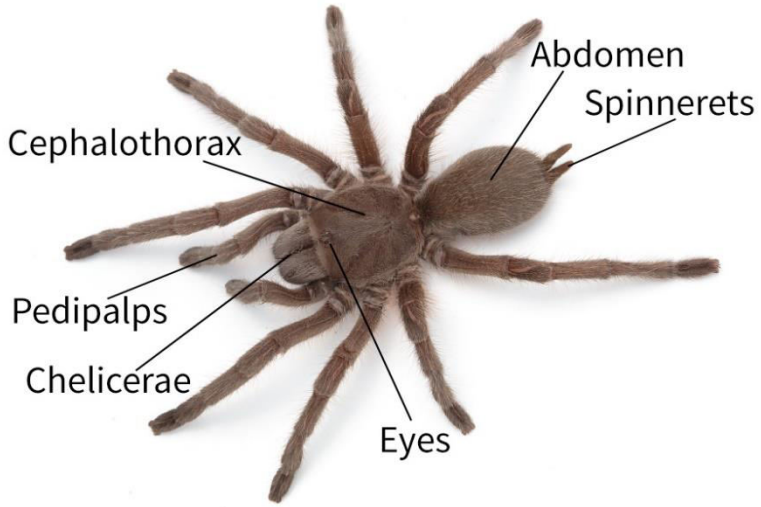
Examples of Instinctive Behaviors



Web making in spiders _ an examples of instinctive or innate behaviour

- A spider web **or cobweb** (from the archaic word coppe, meaning "spider") is a structure created by a spider out of proteinaceous spider silk extruded from its spinnerets, generally meant to catch its prey.

Web making in spiders



- spiders produce silk threads to build their webs.
- silk is produced in silk glands with the help of the spider's spinnerets.
- **Spinnerets** are special organs that allow the spider to decide what type of thread it needs for the web.
- The silk threads can be thick or thin, dry or sticky, beaded or smooth.
- The threads a spider uses to construct its web begin as liquid, but they dry quickly in the air.
- Making webs is instinctive for spiders, which means nobody has to teach them how to do it. They are born knowing how.

Web making in spiders



Process of web making

- When a spider begins a web, it releases a silk thread. It anchors the thread to some object such as a branch.
- As the spider moves back and forth, it adds more threads, strengthening the web and creating a pattern.
- Lines that go from the center of the web outward are called **"radial lines“.** They support the web.
- Threads that go around and around the web are called "**orb lines."**

Web making in spiders

- The main reason spiders spin webs is to catch the prey. When an insect, such as a fly, flies into a spider's web, it gets stuck on the sticky threads.
- When a spider catches prey in the sticky strands of its web, it approaches the trapped insect and uses its fangs to inject venom. The venom either kills or paralyzes the prey.

Animal Behavior

Nest building in birds

Examples of Instinctive Behaviors









Nest building in birds_ an examples of instinctive or innate behaviour

- A bird nest is the spot in which a bird lays and incubates its eggs and raises its young.
- For most birds, when a male and female form a breeding pair nest building activities will follow.
- Nest building is an instinctive behaviour as birds have innate ability to build nests.

Nest building in birds

Nest Building Behaviour

- A **nest site** is first selected and nest building begins.
- This can involve both birds as a pair or either sex alone.
- Different species of birds build different nests. Some nests are very simple constructions, whilst others are more complex.

Nest building in birds

- Birds usually only build nests for holding their eggs and rearing chicks.

Birds build nests to:

- To lay their eggs in and keep them safe
- To incubate their eggs
- To raise their chicks
- To protect their eggs and chicks from predators and adverse weather conditions

Nest building in birds

Nests are built in:

- hedges, bushes and trees
- directly on the ground
- Holes, trees or reeds by rivers and lakes
- tunnels in riverbanks and cliffs
- on cliff ledges

Nest building in birds



Weaver Bird's Nest



Crow's Nest



Magpie's Nest

- Sparrow made its nest from stalks of grass.
- The Reed Warbler such as Weaver Bird; weaves its nest and attaches it to reed stems.
- Large birds such as Rooks, Magpies and Crows make very simple flat nests out of twigs. These nests are called platform nests.

Animal Behavior

Innate Behaviour of three-spined stickleback fish

Innate Behaviour of three-spined stickleback fish



- Sticklebacks exhibit stereotyped instinctive behaviour such as migration, nest building, territoriality, aggression, courtship and parental care.
- They are a good example of male dominating and nesting behaviour too.
- Normally breed in late March and early August.
- Usually live in schools.
- But at the beginning of breeding season male isolate itself.

Innate Behaviour of three-spined stickleback fish

The reproductive behaviour in three-spined stickleback fish

A sexually mature stickleback fish male will have a red coloured belly to attract the female. A mature stickleback fish migrates to warm shallow waters.

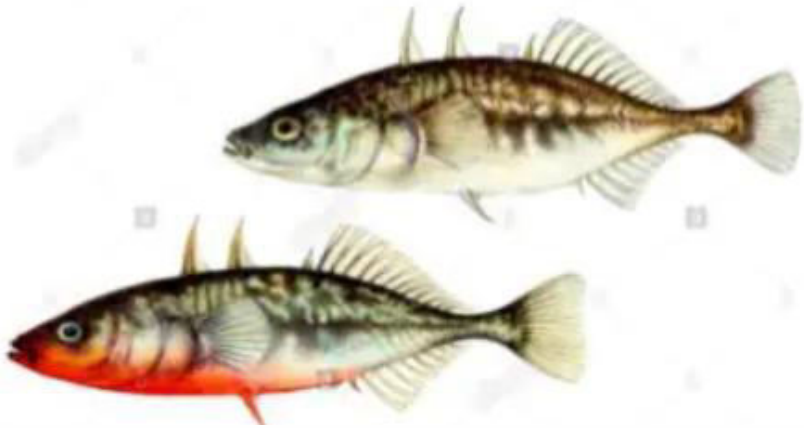
Migration is an instinctive behaviour triggered by the increase in gonadotropic hormones which is in turn controlled by environmental factors like temperature and photoperiod.

Innate Behaviour of three-spined stickleback fish

On reaching shallow waters the fish **builds a nest** which is open at both ends with the help of plants and weeds.

The nest building behaviour is triggered by the presence of green vegetation in the water. Thus he acquires a small territory.

Innate Behaviour of three-spined stickleback fish



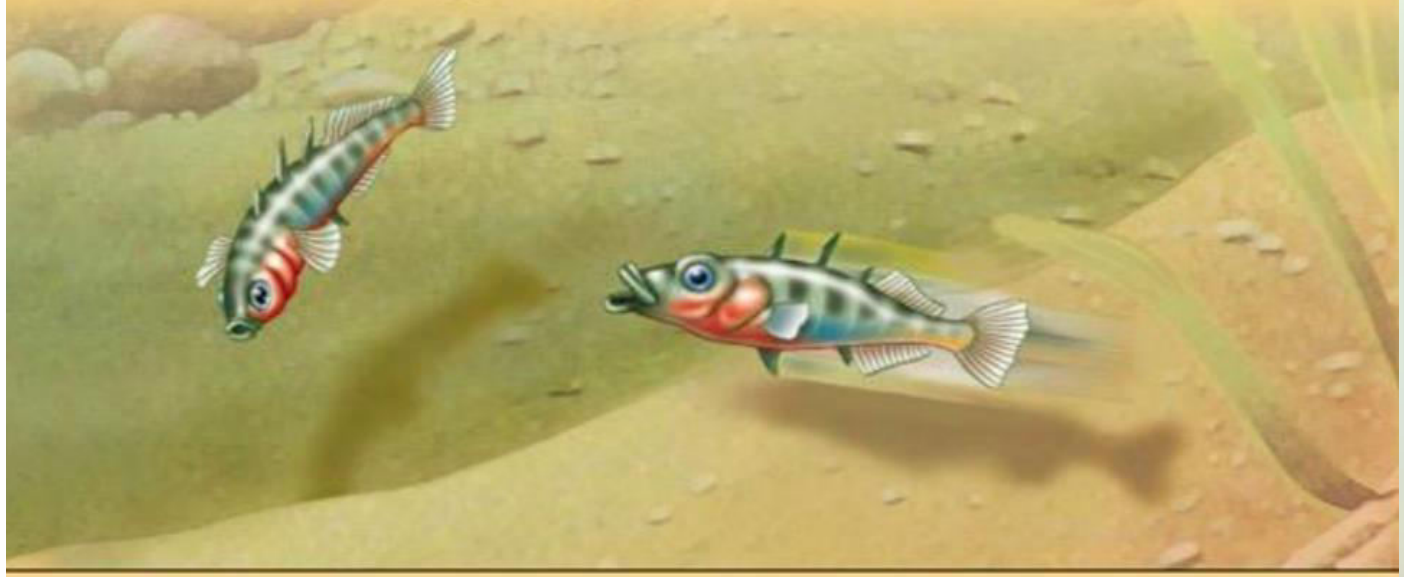
In the early spring.the male stickle backs undergoes quite a dramatic transformation.His throat and belly becomes red-orange,his eyes turns bright blue and silvery scales appears on his back.

It exhibits **territoriality** which means it defends the area around the nest from other male fishes that are also in search of suitable nesting grounds.

The fish will becomes aggressive and exhibits threat posture by facing downwards. The sign stimulus for territorial and aggressive behaviour is the **red belly** of other male fish.

Innate Behaviour of three-spined stickleback fish

BEHAVIOR: A male stickleback fish attacks other male sticklebacks that invade its nesting territory.



PROXIMATE CAUSE: The red belly of the intruding male acts as a sign stimulus that releases aggression in a male stickleback.

ULTIMATE CAUSE: By chasing away other male sticklebacks, a male decreases the chance that eggs laid in his nesting territory will be fertilized by another male.

Innate Behaviour of three-spined stickleback fish

The presence of a female triggers the **courtship behaviour** in the male because of its swollen belly that is full of eggs.

The male performs a ziz-zag dance which will attracts the female. During his dance the male always exposes his red belly to the female.





Innate Behaviour of three-spined stickleback fish



The male leads the female into the nest to releases its ova that are immediately fertilized by the male. After this the female is driven out and the male may lure other females to its nest.

The presence of eggs in the nest triggers **parental care in the male fishes** and it will begin to fan the eggs with its fins and young once are taken care by the males.

Innate Behaviour of three-spined stickleback fish

Hence various stereotyped instinctive behaviour in stickleback fishes are:

- Migration
- nest building
- Territoriality
- Aggression
- courtship and
- parental care

Innate Behaviour of three-spined stickleback fish



Animal Behavior

Swimming in dolphins and other aquatic species

Swimming in dolphins and other aquatic species



- Dolphins are marine mammals closely related to whales and porpoises.
- There are almost forty species of dolphins.
- Dolphins are among the intelligent animals, and their often friendly appearance and seemingly playful attitude have made them popular in human culture.

Swimming in dolphins and other aquatic species



Physical Features

- They have a pointy snout that makes them look like they are smiling.
- They are carnivores.
- Dolphins have a blowhole at the top of their heads that they use to breathe.
- They have upper and lower jaws with small teeth.
- Dolphins have a dorsal fin which keeps them steady while they are swimming and two tail flukes for moving up and down.

Swimming in dolphins and other aquatic species

HABITAT

Dolphins are found all over the world; generally in shallow cold waters as well as the warm tropical waters

Swimming in dolphins and other aquatic species

BEHAVIOR

- They eat between six and fifteen kg. of food per day.
- They swallow their food whole even though they have so many teeth.
- When bottlenose dolphins are ready to eat, they send out a scout to find the food.
- When dolphins are looking for a mate, it’s usually the female who looks for the male.

Swimming in dolphins and other aquatic species

Behavioral Adaptation

Dolphins work together to find food taking turns to feed while the other dolphins keep the fish herded.

Swimming in dolphins and other aquatic species

Innate Behaviour in Dolphins

Dolphins may have lots of learned behaviour but some of the innate behaviour also present in them:

- Know how to swim by birth
- Leaping from water
- Preying and eating food
- communicate with each other using a wide variety of vocalizations

Swimming in dolphins and other aquatic species

Innate Behaviour in Dolphins

- Observations indicate that dolphins undergo daily cycles of activity, which include feeding, socializing, traveling, and resting. All these are innate or instinctive behaviours of dolphins.
- Bottlenose dolphins show aggression and establish and maintain dominance through posturing, biting, chasing, jaw clapping, smacking their tails on the water, emitting bubble clouds from their blowholes, and impact through body slamming in the most severe cases.

End

Animal Behavior

Animal Orientation, Kinesis and Taxis

Animal Orientation, Kinesis and Taxis

Animal Orientation

"Ability of an animal to determine its position in space and move predictably with respect to specific stimuli"

Animal Orientation, Kinesis and Taxis

Animal Orientation: Examples

- Standing up after falling down (orientation against gravity)
- Orienting towards incoming light
- Acoustic orientation
- Olfactory orientation

Animal Orientation, Kinesis and Taxis

Sensory Motor Basis of Orientation

- Animal orientation is a complex process
- Requires integration of sensory input and coordination of motor output

Animal Orientation, Kinesis and Taxis

Input-Output Integration

Integration includes:

- Receiving information through sensory receptor neurons
- Processing and correlating it in the CNS
- Forming a response through muscles that cause orienting movement

Animal Orientation, Kinesis and Taxis

Types of Orienting Movements

- Kinesis
- Taxis

Kinesis

"a change in activity or turning rate in response to a change in location"

Animal Orientation, Kinesis and Taxis

Kinesis—Example

- Kinesis in sow bugs in response to variation in humidity
- Move more rapidly in dry areas and become slow in humid areas
- They survive better in moist areas
- Rapid movement in dry areas to reach moist one is significant for survival

Animal Orientation, Kinesis and Taxis

Taxis

"An oriented movement toward (positive taxis) or away (negative taxis) from some stimulus"

Animal Orientation, Kinesis and Taxis

END

Taxis—Examples

- River fishes (e.g. trout) swim or orient in an upstream direction (toward the current)
- Helps keep the fish from being swept away
- Also keeps facing the direction from which food may come
- Cockroaches move away from light (-ve phototaxis)

Animal Behavior

Types of Kinesis

Types of Kinesis

Kinesis

- Kinesis is a type of locomotory behaviour in relation to the source of stimulus.
- The animal responds to the variation in the intensity of the stimulus and not the source of direction of the stimulus.
- To respond to such stimulus the animal only requires sense organs sensitive to variation in stimulus intensity.

Types of Kinesis

There are two types of kinesis

1. Orthokinesis
2. Klinokinesis

Types of Kinesis

Ortho kinesis

It is a response that involves changes in the speed of movement of the whole body in response to stimuli like humidity, preserve and diffused light.

Types of Kinesis



Examples:

- Wood louse, *Porcellerio scaber,* tend to lose water from their body fairly rapidly when exposed to low humidity.
- When kept at the junction in a choice chamber where one side has high humidity and the other half has low humidity.
- It begin to move and the speed of movement and the rate of turning is greatest in the driest part of the chamber.
- This increased and apparently random movement is an attempt by the animal to remain in the most favorable environment.

Types of Kinesis

Klinokinesis

The speed of locomotion remains constant but the rate at which the animal changes direction depends on the intensity of the stimulus.

Types of Kinesis



Examples

- A planaria changes its direction ever so often as it crawls. If the light intensity above the animal is increased it changes direction more frequently but moves at the same speed .
- The increase in rate of turning falls after sometime but increases again with a further increase in the intensity of light.

End

Animal Behavior

Taxis and its types

Taxis and its types

Taxis

Taxis is the simplest type of innate or stereotyped behaviour. It is an orientation of an animal (directed either towards or away) in response to the source of stimulus.

Taxis and its types

Types

- **Positive taxis**

If the orientation is towards the stimulus it is called as positive taxis

- **Negative taxis**

If the orientation is away from the stimulus it is known as negative taxis

Taxis and its types

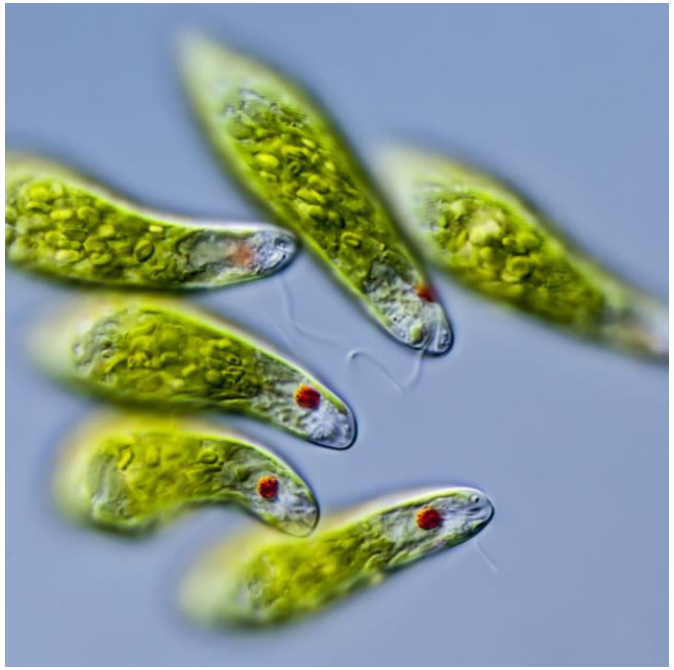
Taxes are usually named after the stimuli. Hence there are:

- Phototaxis
- Chemotaxis
- Thermotaxis
- Geotaxis

Taxis and its types

Examples

- A plant or animal that move towards light shows positive phototaxis.
    - The protozoan, Euglena response to a variety of stimuli and is very sensitive to light. It swims towards light; hence it shows positive phototaxis.



Taxis and its types



End

Examples

- Certain burrowing animals more away from the source of light exhibits negative phototaxis.
  - Maggots that are about to pupate moves away from light source towards a dark location hence they show negative phototaxis.

Animal Behavior

Reflexes and its advantages

Reflexes and its advantages

Reflexes

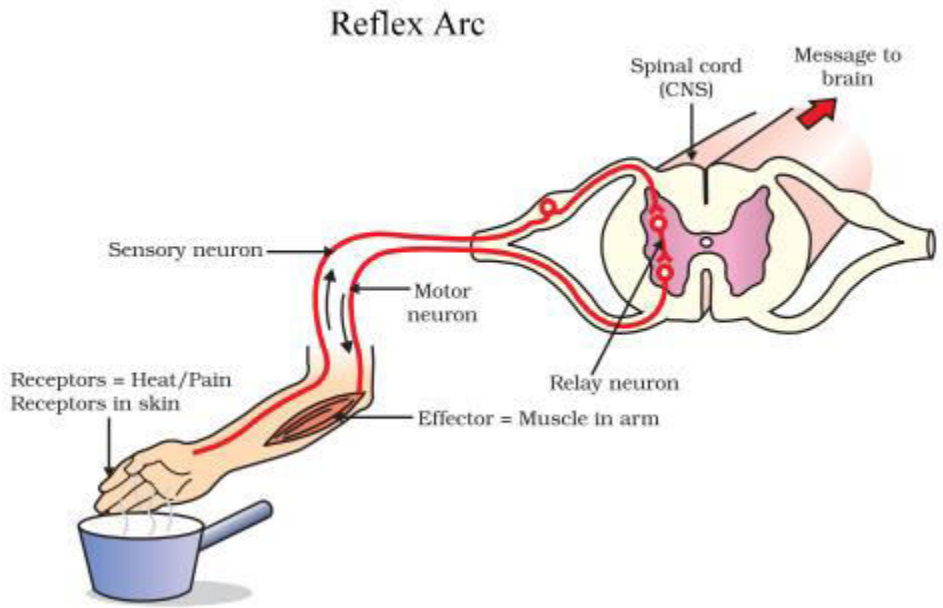
- A simple movement of a part of the animal in response to a stimulus is called **reflex**.
- It is a quick, innate and immediate response of a part of the body to an external or internal stimulus which has great adaptive and survival value to the organism.
- Reflexes are inherited and unlearned behaviour found in all members of the species.

Reflexes and its advantages

Examples

- Kneejerk
- Constrict of pupil of eye in the bright light
- Blinking of eye
- Peristalsis and coughing in man
- Flight in birds
- Web spinning in spiders

Reflexes and its advantages



A **reflex action** requires a **reflex arc** which consists of:

- a sensory organ (receptor),
- a sensory nerve (afferent nerve),
- the spinal cord or brain,
- an intermediate motor or efferent nerve
- a motor organ or effector

Reflexes and its advantages

A reflex action thus involves the sense organs and nervous system.

Steps

1. The stimulus is received by the receptor or sense organ which sets up a sensory impulse.
2. This impulse is transmitted to the central nervous system (brain and spinal cord) through the intermediate nerve.
3. This motor impulse is transmitted to the effector or motor organ, muscles and glands which either contract or secrete in response to the stimulus.

Many pattern of animal behaviour are complex combinations of simple reflex.

Reflexes and its advantages

Advantages of reflex

- Enables the animal to respond immediately to harmful stimuli hence it has great adaptive and survival value
- Since many of the reflex actions are controlled by the spinal cord, it relieves the brain from too much work.

End

Animal Behavior

Fixed Action Patterns

Fixed Action Patterns

Fixed Action Patterns (Instincts)

- It is a sequence of unlearned, innate behaviour that is unchangeable.
- Once initiated; it is usually carried to completion.
- It is hardwired, however, not purely genetic, may improve with experience.
- FAP is triggered by an external sensory stimulus known as “sign stimulus”.

Fixed Action Patterns

Konrad Lorenz is the founder of the concept.

Lorenz identified **six characteristics** of fixed action patterns. FAPs are:

- Stereotyped
- Complex
- Species-characteristic
- Released
- Triggered
- Independent of experience

Fixed Action Patterns

Properties of Fixed Action Patterns

- Are relatively complex motor acts
- Are all-or-none responses i.e. once initiated, continue to completion even if the stimulus is removed

Fixed Action Patterns

Properties of Fixed Action Patterns

- All members of a species perform a given fixed action pattern identically
- Patterns are inherited genetically
- Are exhibited perfectly without prior experience

Fixed Action Patterns

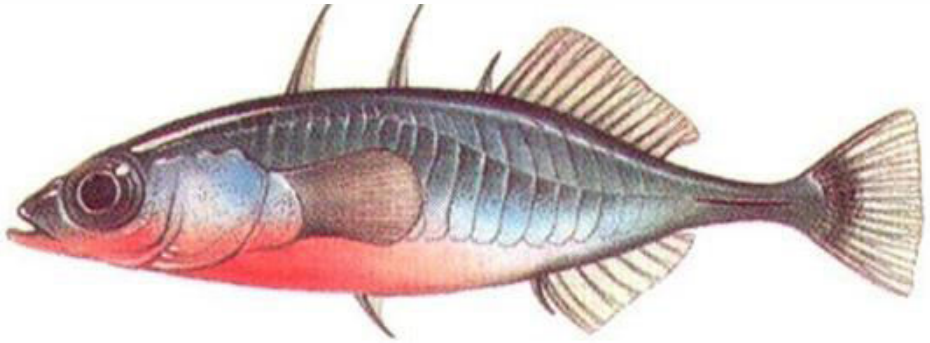
Properties of Fixed Action Patterns

- Stimulus threshold may vary with the physiologic state of animal. For example:
  - An animal has copulated once to an arousing stimulus
  - It cannot be induced to copulate again without intense stimulation

Animal Behavior

Fixed Action Patterns 2

Fixed Action Patterns



Example

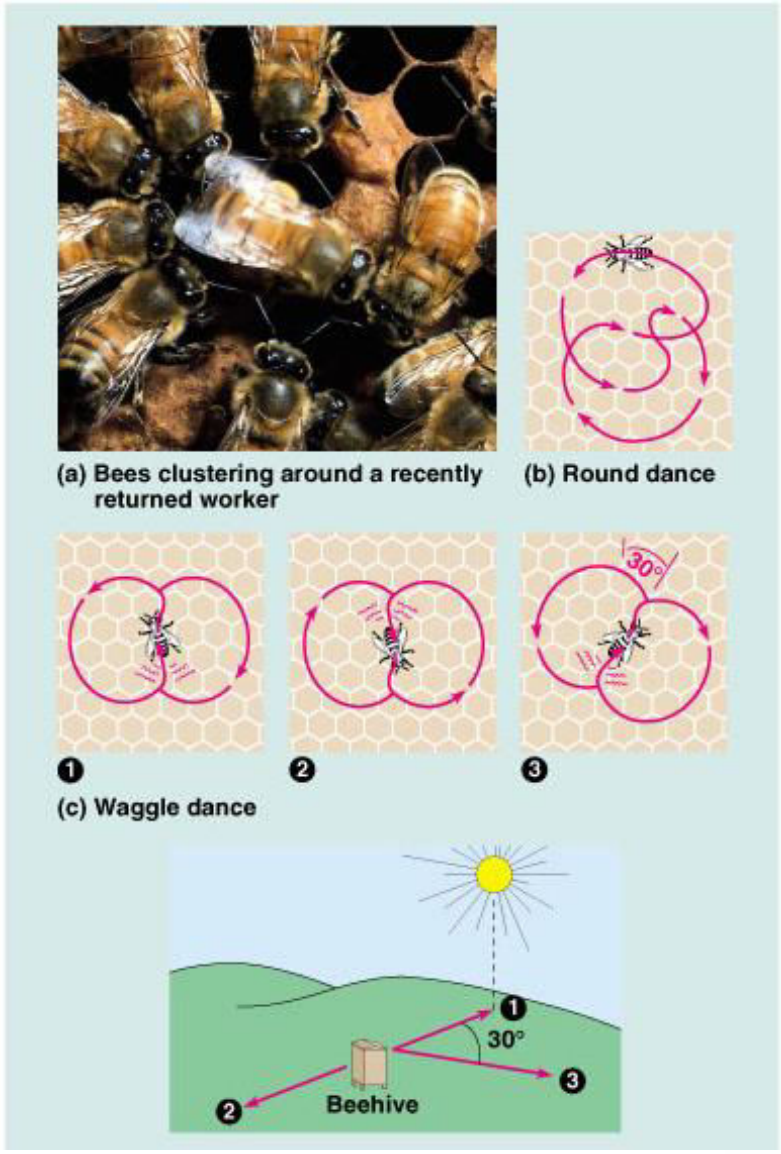
Male three-spined stickleback fish

- Male three-spined stickleback fish have red bellies
- A male perceives other males with red bellies, coming to their nesting territories as invaders
- Shows off aggressive attacking behavior

Fixed Action Patterns

- It attacks anything with red underside even unrealistic models and objects
- Red underside is a sign stimulus
- Aggressive attacking behavior is an instinct (fixed action pattern)

Fixed Action Patterns



Example

Communication in bees by Karl von Frisch

- A special dance performed by a worker bee called **“waggle dance”**
- Workers returned to the honey comb with pollen or nectar,
- Dances constitute a language that “tells” other workers where the food is.

Fixed Action Patterns

- Dance tells the direction and distance of the food source
- Worker bee uses this dance to recruit and direct other workers in gathering pollen and nectar.
- In 1973, Karl von Frisch was awarded the Nobel Prize for his research on the honey bee waggle dance.

Fixed Action Patterns



Example

Herring gull experiment by Niko Tinbergen

- Releasers (red spot on the bill) initiate behaviour automatically.
- From birth, herring gull chicks peck at a red spot on their parent's bill in order to get food.

Fixed Action Patterns

- They will also readily peck at red dots on the end of a wagging pencil, and a wagging model with the red spot on its head. Herring gull chicks respond poorly to stationary models and models with no spot.

End

Animal behaviour

Learned behaviour or
Acquired behaviour

Types of Animal behaviour

There are 2 basic types of behaviour

1. Innate behaviors or Inherent behaviour
2. Learned behaviors or acquired behaviour

1. Nature/innate

Instinct or genes determine behaviour

2. Nurture/learned

Experience and learning influence behaviour

Learned behaviour or Acquired behaviour

2. **Learned behaviors**

- modified by experience
- variable, changeable
- flexible with a complex & changing environment

Learned behaviour or Acquired behaviour

Learned (acquired) behaviour

Thorpe defined learning as a long lasting adaptive change in behaviour resulting from past experience; hence it is acquired during the life time of an individual.

Learned behaviour or Acquired behaviour

Characteristics of learned behaviour

1. Acquired during the life of an organism due to constant experience
2. Experience dependent and can be modified through experience
3. Learning is flexible
4. Learning behaviour differs from individual to individual, to individual among the same species hence not species specific

Learned behaviour or Acquired behaviour

Characteristics of learned behaviour

5. Learned behaviour is certainly not inherited, though the ability to learn is almost certainly inherited as it is dependent on the development of the nervous system of the organism which is inherited.

6. All organisms have the ability to learn at least to some extend.

7. However learning in humans surpasses all other animals. Almost everything we do has been learned. No other species surpasses the humans in their amount and range of information that is acquired through learning

Learned behaviour or Acquired behaviour

Types of learning

i. Habituation
ii. Imprinting
iii. Classical conditioning
iv. Trial and error learning
v. Latent learning
vi. Insight
vii. Reasoning
viii. Cognition

End

Animal Behavior

Habituation and Imprinting

Habituation and Imprinting

Learning Behaviors

- Behaviors that are variable with experience

Examples

- Habituation
- Imprinting

Habituation and Imprinting

Habituation

- One of the simplest forms of learning
- An animal learns to ignore a repeated, irrelevant stimulus

Definition

"A loss of responsiveness to stimuli that convey little or no new information"

Habituation and Imprinting

Habituation—Examples

- Many mammals and birds recognize alarm calls of members of their species
- If these calls prove fake repetitively, they stop responding ("cry-wolf" effect)

Habituation and Imprinting

Habituation—Examples

- A threatened squirrel gives signals
- Others recognize signal and go to nearest refuge
- An individual who gives many false alarms is also recognized
- Its signals are ignored

Habituation and Imprinting

Significance of Habituation

- An animal conserves energy and time by habituation to unimportant stimuli
- Energy and time are better spent on other important functions

Habituation and Imprinting

Imprinting

"Formation of a long-lasting behavioral attachment to a particular individual or object at a specific stage in life"

Habituation and Imprinting

Imprinting

- A type of learning behavior
- A very young animal develops attachment with the very first object it has visual, auditory or tactile experience
- It thereafter follows that object

Habituation and Imprinting

Sensitive or Critical Period

- A specific sensitive (critical) period in which imprinting occurs
- A limited developmental phase for certain learned behaviors
- Young imprint on their parent to learn basic behaviors of species
- Parent learns to recognize its offspring

Habituation and Imprinting

Imprinting—Example

- Seagulls
- Sensitive period for a parent to bond with its young lasts one to two days
- If bonding does not occur, parent will not care for the infant, leading to death of the offspring

Habituation and Imprinting

Imprinting—Example

- Classic experiment of Konard Lorenz with geese



Habituation and Imprinting

Imprinting—Example

- Konard Lorenz made goslings to imprint on him
- Goslings followed him as though he was their mother



END

Animal behaviour

Examples of Habituation and Imprinting

Examples of Habituation and Imprinting

Examples of Habituation

- Lack of continued response to strong odour is a common example of sensory habitation. Sensory system may stop after sending signals to the brain in response to a continuous repeated stimulus.

Examples of Habituation and Imprinting



Examples of Habituation

- If any disturbance is created in the spider web, the spider will response to it immediately but finding nothing it will return.
- If this stimulus is repeated several times, spider will gradually stop responding as the stimulus was of no important consequence to the spider and hence it will ignore the stimulus.
- However the spider will respond if some other end of the web is disturbed.

Examples of Habituation and Imprinting



Examples of Habituation

- Prairie dogs give alarm calls when mammals, large birds or snakes approach which warns others of the presence of these predators. These alarm calls facilitate escape into their burrows.
- However, if their burrows are located near trials used by humans, the given alarm calls every time a person walks by would be a wasted effort. Thus habituation to humans is an important adaptation to these animals.

Examples of Habituation and Imprinting



Examples of Imprinting

- When an animal learns to distinguish what an appropriate mate looks like to avoid inbreeding. It is observed in goats, zebra finches, and pandas.

Examples of Habituation and Imprinting



Examples of Imprinting

- The peregrine falcon has also been known to imprint on specific structures for their breeding grounds such as cliff sides and bridges and thus will favour that location for breeding.

Examples of Habituation and Imprinting

Examples of Imprinting

- Angelo d'Arrigo extended the technique used by Konrad Lorenz.
- He noted that the flight of hang-glider is very similar to the flight patterns of migratory birds.
- Both use updrafts of thermal currents to gain altitude that then permits soaring flight over distance.

Examples of Habituation and Imprinting

Examples of Imprinting

- He used this method to reintroduce threatened species of raptors.
- As birds hatched in captivity have no mentor birds to teach them traditional migratory routes, D'Arrigo hatched chicks under the wing of his glider and they imprinted on him. Then, he taught the fledglings to fly and to hunt.
- The young birds followed him not only on the ground (as with Lorenz) but also in the air as he took the path of various migratory routes.

Animal Behavior

Associative Learning

Associative Learning

Associative Learning

"A learning that involves making associations between experiences"

Example

- A mouse bites a brightly colored caterpillar
- It tastes bad
- Mouse avoids attacking insects of similar color in future.

Associative Learning

Associative Learning

- Ability to associate one environmental feature (such as a color) with another (such as a foul taste)

Types

- Classical conditioning
- Operant conditioning

Associative Learning

Classical conditioning

- An arbitrary stimulus becomes associated with a particular outcome

Example

- Experiments of Russian physiologist Ivan Pavlov
- He always rang a bell just before feeding a dog
- Dog conditioned to salivate at the bell's sound alone, anticipating arrival of food

Associative Learning

Operant Conditioning

"An animal learns to associate one of its own behaviors with a reward or punishment and then tends to repeat or avoid that behavior"

- also called trial-and-error learning

Associative Learning

Operant Conditioning: Examples

- A predator learns to avoid certain kinds of prey if associated with painful experience
- An animal may repeat an action that has resulted in a surprise finding of food
- A classic experiment: a rat was trained to obtain food by pressing a lever

END

Animal Behavior

Insight Learning

Insight Learning

Insight Learning

- Insight is immediate and clear learning
- understanding that takes place without overt trial-and-error testing
- Insight occurs when animals recognize relationships (or make novel associations between objects or actions) that can help them solve new problems

Insight Learning

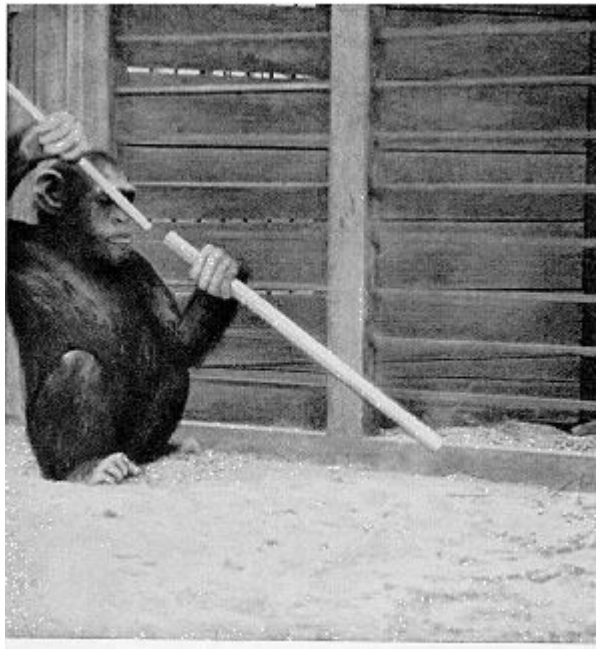
- A type of learning that uses reason, especially to form conclusions, inferences, or judgments, to solve a problem.
- It is based on advanced perceptual abilities such as thought and reasoning

Insight Learning

Kohlar's work on chimpanzees is the experimental proof of insight learning.

- During experiment Köhler placed a banana outside the cage of a hungry chimpanzee
- two sticks was given to Chimpanzee, but too short for pulling in the food but joinable to make a single stick of sufficient length

Insight Learning



- First; chimpanzee tried to use only a stick to reach banana
- Then it tried to use one stick to push the other along to touch the banana.
- Later, it accidentally joined the sticks and observed the result and used it to retrieve the banana.
- In repeated experiments, the chimpanzee joined the two sticks and took the banana immediately.

Animal Behavior

Concept of Animal Cognition

Concept of Animal Cognition

Animal Cognition

- ability of an animal's nervous system to perceive, store, process, and use information gathered by sensory receptors
- encompasses the mental capacities of non-human animals

Concept of Animal Cognition

Animal Cognition

- Animals process information from eyes, ears, and other sensory organs to perceive the environment.
- Problem solving can be learned by observing the behavior of other animals.

Concept of Animal Cognition

Researchers have examined animal cognition in:

- mammals (primates, cetaceans, elephants, dogs, cats, horses, cattle, raccoons and rodents),
- birds (including parrots, fowl and pigeons),
- reptiles
- (lizards, snakes, and turtles)
- Fishes
- Invertebrates (cephalopods)

Concept of Animal Cognition



Examples of Animal Cognition

- Chimpanzees learn to join sticks to reach bananas
- Clark's nutcrackers can find food hidden in caches located halfway between particular landmarks.

Concept of Animal Cognition

Examples of Animal Cognition

- Ravens showing sophisticated social cognitive skills including the formation of coalitions, considering visual perspectives of others, and directing conspecifics' attention to external referents

Animal Behavior

Spatial Learning and Cognitive Maps

Spatial Learning and Cognitive Maps

Spatial Learning

“Capacity to establish a memory that reflects environment's spatial structure”

Spatial Learning and Cognitive Maps

Spatial Learning

- Spatial learning enhances an organism's fitness in environment
- Every natural environment has some spatial variation e.g.,
  - Locations of nest sites
  - Hazards
  - Food
  - Prospective mates

Spatial Learning and Cognitive Maps

Spatial Learning—Example

- Behavior of a female digger wasp
- Nests in small burrows dug into sand mounds
- A wasp covers the entrance of her nest with sand when going on hunting

Spatial Learning and Cognitive Maps

Spatial Learning—Example

- Upon return, she flies directly to her hidden nest
- Despite hundreds of other burrows in the area
- Locates her nest by memorizing its position relative to visible landmarks (location indicators)

Spatial Learning and Cognitive Maps

Spatial Learning—Example

- Proof of visual learning of object maps
- Experimentally disturbing the landmarks confuses the wasp

Spatial Learning and Cognitive Maps

Cognitive Maps

"A representation in the nervous system of the spatial relationships between objects in an animal's surroundings"

- It involves relating landmark positions to one another

Spatial Learning and Cognitive Maps

Cognitive Maps

- Cognitive maps follow abstract geometric rule—identifying halfway point between landmarks
- Reduce the amount of detail required to remember an object's location

Spatial Learning and Cognitive Maps

Cognitive Maps

- Advanced than relying solely on moving from landmark to landmark
- Using cognitive maps makes navigation more flexible and efficient

Spatial Learning and Cognitive Maps

Cognitive Map—Example

- Clark's nutcracker birds
- Store thousands of pine seeds in fall at thousands of hiding places
- Caches—distributed over a large area (35 km^2)
- During winter, they relocate their caches

END

Animal Behavior

Cognition and Problem Solving

Cognition and Problem Solving

Cognition

"The process of intellect, represented by awareness, understanding, reasoning, recalling memories and judgments"

Cognition and Problem Solving

Cognition

- Most complex form of learning
- Characteristically exhibited by primates
- Also shown by:
  - Many mammals
  - Cephalopods
  - Insects

Cognition and Problem Solving

Problem Solving

"A cognitive activity of devising a method to proceed from one state to another in the face of apparent obstacles"

- Depends on information processing ability of nervous system

Cognition and Problem Solving

Problem Solving Behavior

- Problem solving behavior is highly developed in:
- Some mammals (primates and dolphins)
- Some birds (ravens, crows, jays)

Cognition and Problem Solving

Problem Solving Behavior

- Animals may learn by observing behavior of others
- Young wild chimpanzees, copying experienced chimpanzees & learn how to crack oil palm nuts with two stones

Cognition and Problem Solving

Problem Solving: Example

- A chimpanzee placed in a room
- Several boxes on the floor
- A banana hung high out of reach
- Chimp can “size up" by stacking the boxes
- Enabling to reach food

END

Animal Behavior

Ecology and Genetics to maintain animal behaviour

Ecology and Genetics to maintain animal behaviour

Environmental Factors in Behavior Development

- Genes of an individual provides instructions for behavior development
- Expression of genes is influenced by many environmental factors
- Animal behavior develops by complex interactions between genetic and environmental factors

Ecology and Genetics to maintain animal behaviour

Environmental Factors in Behavior Development

- Environment of fertilized egg
- Animal's diet
- Animals social interactions
- Environmental conditions in animal’s surrounding

Ecology and Genetics to maintain animal behaviour

Study of Environment's Effect on Gene Expression

- Two methods to determine extent of environmental effects on expression of behavioral genes:
  - Cross-fostering study
  - Twin Studies

Ecology and Genetics to maintain animal behaviour

Cross-Fostering Study

- Young of one species are nurtured by adults of another species
- Extent of change in offspring's behavior is determined
- Changes are a measure of how social and physical environment influences behavior

Ecology and Genetics to maintain animal behaviour

Cross-Fostering Study

- Males of certain mouse species have behavioral differences that are well suited for cross-fostering studies

Ecology and Genetics to maintain animal behaviour

Twin Studies

- Influence of genetics and environment on human behavior, is explored by twin studies
- Behaviors of identical twins are compared which are:
  - Raised apart
  - Raised in the same household

Ecology and Genetics to maintain animal behaviour

Twin Studies

- Twin studies are carried out for human behavioral disorders:
  - Schizophrenia
  - Anxiety disorders
  - Alcoholism
- Reveal effect of different environments on the expression of disorder
- Vary 20-80 % in separate environments

Ecology and Genetics to maintain animal behaviour

Twin Studies

- Significant variation in development of these disorders in humans prove contribution of both environmental and genetic factors

END

Animal Behavior

Animal Signals and Communication

Animal Signals and Communication

Why study animal communication?

- Central to the study of animal behavior
- Tool for elucidating general evolutionary principles

– Taxonomy, reproductive isolation, speciation, natural selection

Animal Signals and Communication

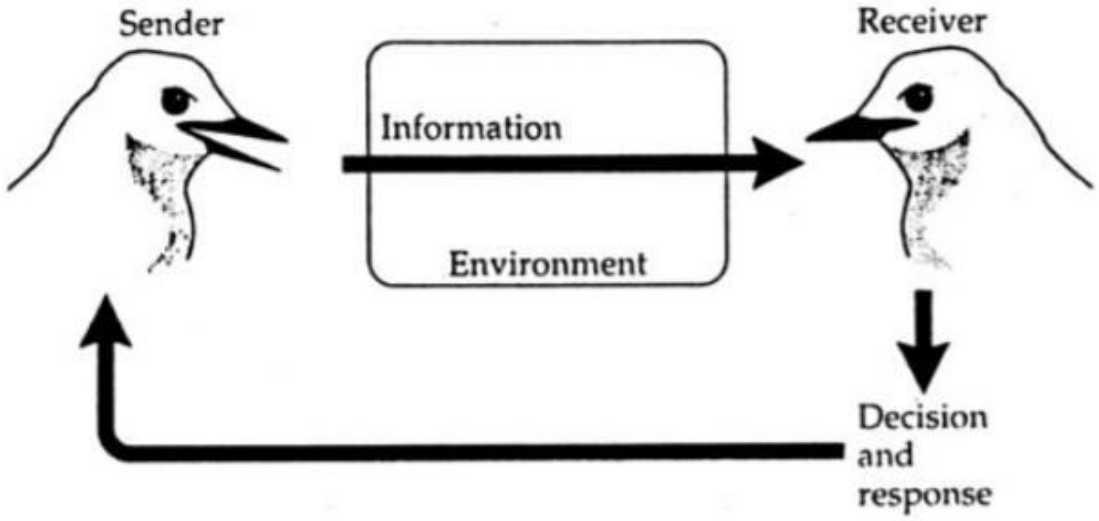
Why study animal communication?

- Has practical and conservation importance
  - – Censusing populations and communities
  - – Insect pest control via mate attraction signals
  - – Welfare of captive and wild animals

Animal Signals and Communication

- There is a sender and a receiver in communication.
- Study of animal communication is called **Zoosemiotics.**
- Classical ethological view was developed by **Niko Tinbergen.**

Animal Signals and Communication



Communication

"Communication is the transfer of information via signals sent in a channel between **sender** and a **receiver** causing some kind of change in the animal that gets the information.

Animal Signals and Communication

2 Types of Communication

- **Intraspecific Communication**

Communication within a single species, e.g., honeybee dancing

- **Interspecific Communication**

Communication among different species

Prey to predator e.g., warning colouration in wasps

Predator to prey e.g., Angler fish

Human/animal communication during domestication

Conti…

Animal Behavior

Animal Signals and Communication

Animal Signals and Communication

Animal Signals

"A stimulus transmitted from one animal to another"

- Signals may be:
  - Visual
  - Chemical
  - Tactile
  - Auditory

Animal Signals and Communication

Animal Communication

- Transmission and reception of signals
- Essential element of interactions between individuals
- Used to transmit information among individuals

Animal Signals and Communication

Communication Patterns

Correlated with signals:

- Visual communication
- Olfactory communication
- Tactile communication
- Auditory communication

Animal Signals and Communication

Using Form of Communication

- Closely related to an animal's lifestyle and environment

Animal Signals and Communication

Examples—Form of Communication

- Diurnal animals communicate primarily by visual and auditory signals
- In a nocturnal terrestrial mammals visual displays are relatively ineffective
- They use olfactory and auditory signals—work well in dark

Animal Signals and Communication

Example of Animal Communication

- Courtship behavior of fruit fly *Drosophila melanogaster*
- Constitutes a stimulus-response chain
- Response to each stimulus is itself stimulus for the next behavior

Animal Signals and Communication

Courtship Behavior of Drosophila

- A male identifies a female of its species—visual communication
- Male detects chemicals released by female by smell–olfactory or chemical communication
- Male approaches and touches the female with a foreleg—tactile communication

Animal Signals and Communication

Courtship Behavior of Drosophila

- Male extends and vibrates wings: courtship song—auditory communication

Result of Successful Communication

- Female identifies male as member of the same species
- Becomes receptive
- Allows male to attempt copulation

Animal Signals and Communication

Communication for Information Transfer

- Information content varies considerably

Example

- Symbolic language of honeybees
- Share information about the location of food sources

Animal Signals and Communication

Symbolic Language of Honeybees

- A honeybee finds a food source close to the hive (less than 50 m away)
- Returns to its hive
- Moves in tight circles, moving its abdomen from side to side "round dance”
- Motivates other bees to leave the hive
- Follow her for nearby food

Animal Signals and Communication

END

Symbolic Language of Honeybees

- When food source is farther
- “Waggle dance” is performed
- Swings half-circle in one direction, a straight run, a half-circle swing in other direction
- Communicates direction and distance of source

Animal Behavior

Pheromones and Chemical Communication 1

Pheromones and Chemical Communication

Pheromones

- Chemical messengers released by many animals in environment to communicate
- Comprise 2 Greek words: “Pherein”(to transfer) and “Hormon” (to excite)

Pheromones and Chemical Communication

Pheromones

- Spread through air
- Affect the behavior of individuals of a species
- Effective at remarkably low concentrations
- Have particular odors
- Detected by olfactory receptors of receiving animal

Pheromones and Chemical Communication

Chemical Communication

- Chemical communication through pheromones is well developed in:
  - Unicellular organisms
  - Insects
  - Fishes
  - Salamanders
  - Mammals

Pheromones and Chemical Communication



Examples

- **Queen bee** gives off a pheromone that prevents other females in the hive from becoming sexually mature, with the result that only the queen bee mates and lays eggs.
- Bees release an alarm hormone right after stinging a predator.

Pheromones and Chemical Communication



Examples

The **mountain lion** is a solitary animal and it likes to travel alone. A male's territory can be more that 100 square miles. The male mountain lion often kicks up piles of pine needles, dirt or leaves with its hind feet and urinates on the pile to mark its territory.

Pheromones and Chemical Communication



Examples

The **Canada lynx** also leaves a scent post. It will urinate on trees and tree trunks. It may also leave claw marks on trees. Claw marks are a visual sign, but can also be a chemical one. Many animals have scent glands in their claws. When they scratch a tree, they leave a little of their scent behind.

Pheromones and Chemical Communication



Examples

Some animals use scent to identify members of their family. The **mule deer** has scent glands on its hind legs right above its hooves. Young deer will often identify their mothers by sniffing at the glands. When mule deer are in groups, they often sniff each others hind legs. When male mule deer become aggressive towards each other, the hair around the glands will stand up.

Pheromones and Chemical Communication



Examples

Domestic cats have scent glands near their mouths, on their foreheads and at the base of their tails. They use these glands to mark territory.

Pheromones and Chemical Communication

Examples

Many animals use chemical communication or pheromones to communicate. They may leave their scent to mark territory or attract mates. They often use smell to find prey and identify other animals.

Pheromones and Chemical Communication

Examples

Some animals leave **scent posts** to mark their territory. A scent post is something that is marked by an animal with its scent. Scent posts set the boundaries of an animal's territory.

Pheromones and Chemical Communication



Examples

- Some animals leave **scent posts** to mark their territory. A scent post is something that is marked by an animal with its scent. Scent posts set the boundaries of an animal's territory.

- Ants use trail forming or recruitment pheromones that are useful in foraging.

Animal Behavior

Pheromones and Chemical Communication 2

Pheromones and Chemical Communication

Roles of Pheromones

1. Recognize Members of Species

- Unicellular organisms recognize members of their own species
- Many insects identify members of their colony

Pheromones and Chemical Communication

Roles of Pheromones

2. Reproductive Roles

- Pheromones attract members of opposite sex for breeding
- Induce courtship behaviors in members of a species
- Induce spawning behavior in aquatic animals with external fertilization

Pheromones and Chemical Communication

Roles of Pheromones

3. Territorial Behaviors

- Many male mammals mark their territories with pheromones
- It warns other males of their occupied area

4. Insect’s Social Order

- In a honeybee colony, pheromones produced by queen maintain hive's complex social order

Pheromones and Chemical Communication

Roles of Pheromones

5. Alarm Calls

- Members of some species produce pheromones on getting injured
- It warns other members of presence of danger in area

Pheromones and Chemical Communication

Roles of Pheromones

6. Repel Predators

- Pheromones may repel predators
- Example: foul-smelling musk that makes skunks unpalatable to their enemies

Pheromones and Chemical Communication

Advantages of Pheromones as Chemical Signal

- Provide a simple message that can last for hours or days
- Are equally effective during night or day
- May be transported over long distances by air
- Take relatively little energy to produce

Pheromones and Chemical Communication

Disadvantages of Chemical Signals

- Chemical signals are slow to act
- They cannot be changed quickly

END

Animal Behavior

Behavioral Rhythms

Behavioral Rhythms

Biological Clocks

Animals respond rhythmically to cyclic fluctuations in environment e.g.

- Changes in seasons
- Daily light and dark periods

Rhythmic behaviors are controlled by biological clocks

Behavioral Rhythms

Biological Clocks

- Biological clocks regulate daily and seasonal cyclic activities
- Clocks have a molecular genetic basis
- Genes for time sense have been identified in many animals (human beings too)

Behavioral Rhythms

Biological Clocks

Biological clocks may be:

- Circadian clocks
- Circannual clocks
- Lunar clocks

These clocks are influenced by:

- Dark and light periods
- Lunar cycles

Behavioral Rhythms

Circadian Clock and Circadian Rhythms

- Most important biological clock in all animals
- Regulates and coordinates daily behavioral activities
- Its output is a circadian rhythm
- Exhibited as a daily cycle of rest and activity

Behavioral Rhythms

Circadian Clock and Circadian Rhythms

- Clock is synchronized with light and dark cycles of the environment
- Can also maintain rhythmic activity under constant environment conditions (e.g. during hibernation)

Behavioral Rhythms

Circannual Rhythms

- Biological rhythms linked to yearly cycle of seasons
- Span events repeating on 12 month basis
- Correspond to tilt of earth's spin axis resulting in seasonal changes
- Circannual behaviors:
  - Hibernation
  - Migration
  - Reproduction

Behavioral Rhythms

Lunar Cycles and Tidal Rhythms

- Controlled by phases of the moon
- Gravitation pull of moon determines timing of tidal rhythms in marine environments

Behavioral Rhythms

Lunar Cycles and Tidal Rhythms

- Reproductive behaviors of many marine animals are linked to the timing of new or full moon
- This behavior links reproduction to the times of greatest tidal movement

Behavioral Rhythms

Lunar Cycles and Tidal Rhythms

- Tides disperse larvae to deeper waters
- Deeper waters are relatively safer environment to complete early development

END